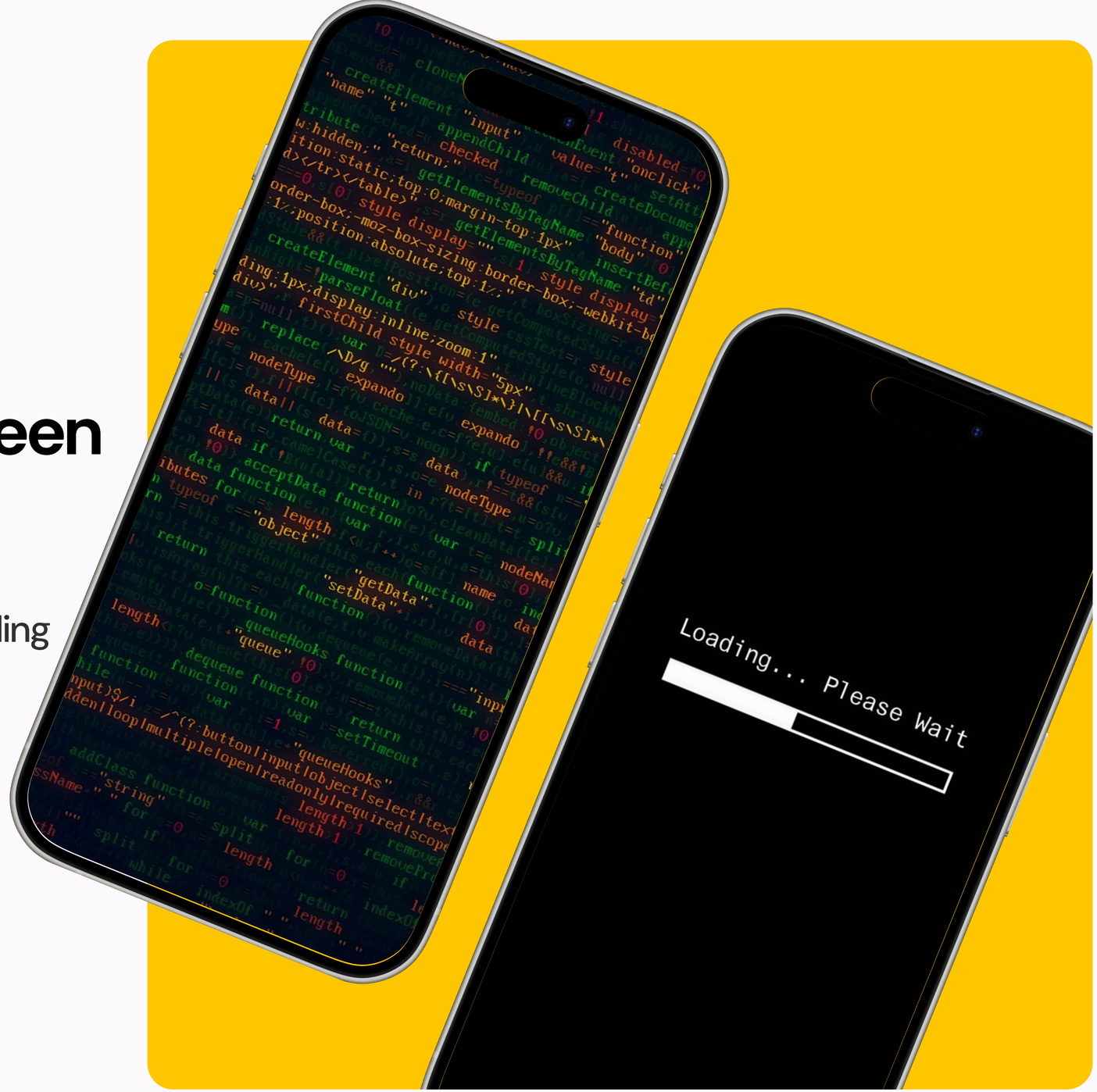
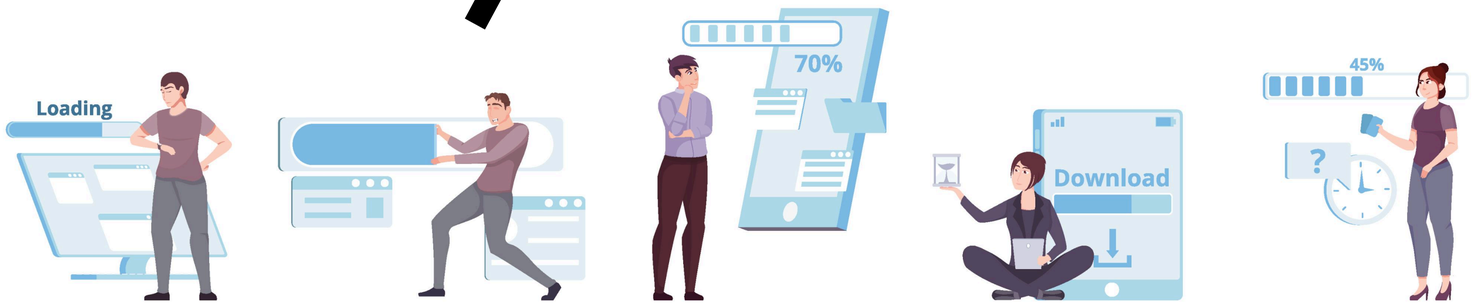
The hidden processes between loading pages

A visual journey into what happens behind every loading screen on websites



• Student: Kristian Hajdari • Project name: Making the invisible visible • Year: 2026

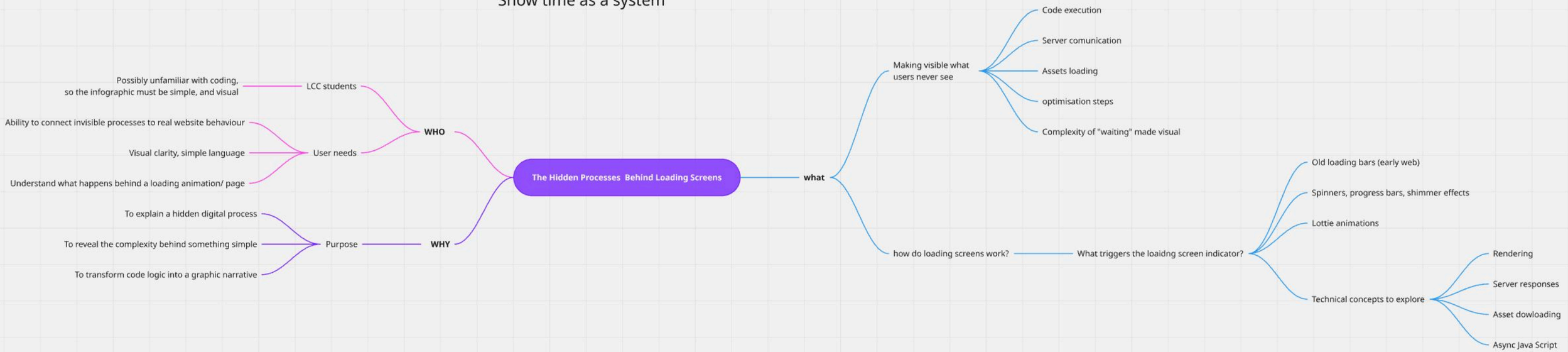
Part 1;



Context & Questions

My goal:

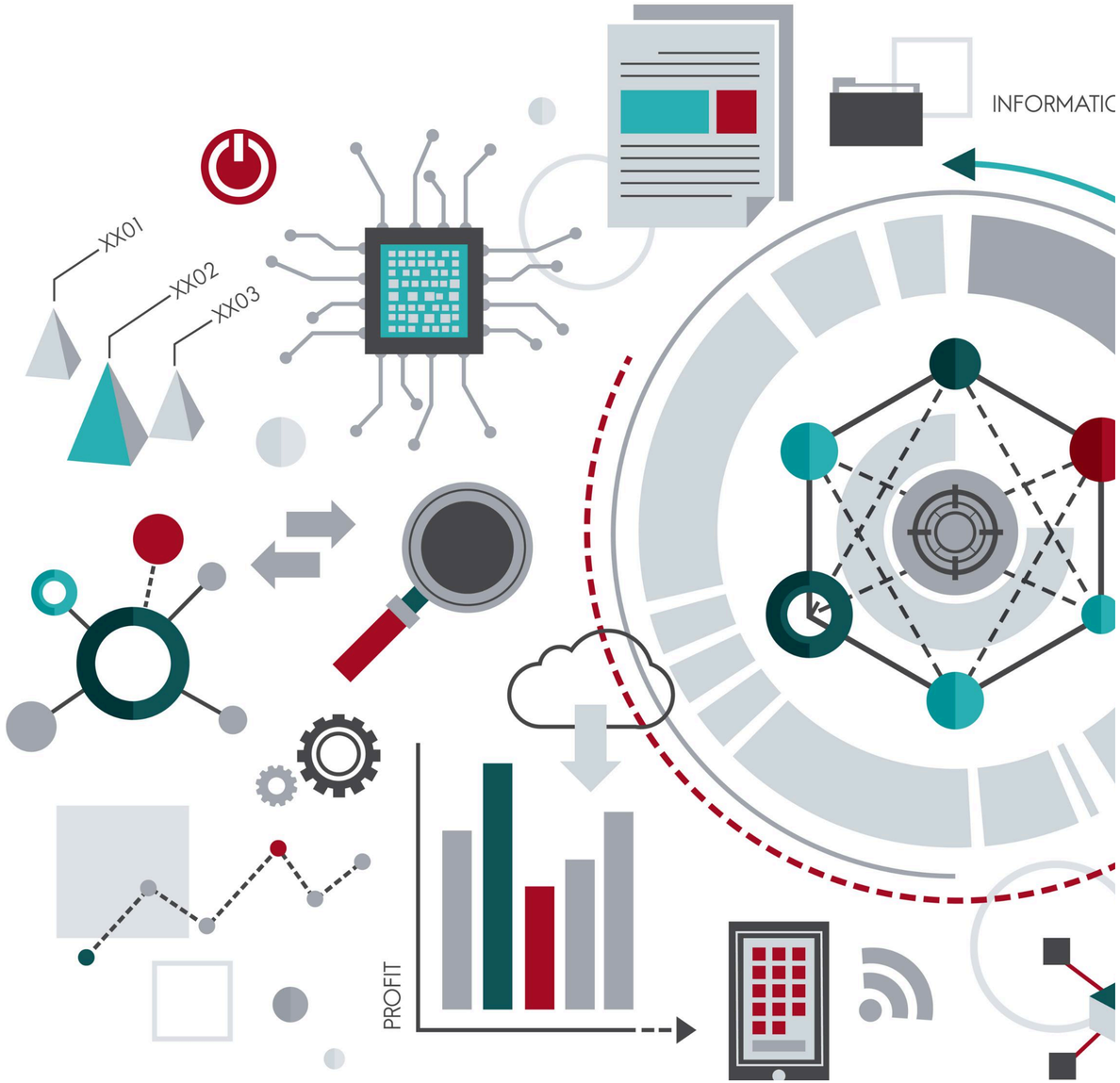
Reveal
Explain
Translate complexity (make it simple)
Make digital process understandable
Show time as a system



The mindmap

My goal

- Reveal what normally stays hidden during a loading moment.
- Explain digital processes.
- Translate complexity into a visual system that anyone can understand.
- Show “waiting time” as an actual sequence of operations, not just a pause.



Who - The audience

LCC Students:

- Probably not familiar with coding.
- Need a simple, visual explanation.
- Need help connecting “invisible processes” to what they experience on a website.



Who - The audience

User needs:

- Visual clarity
- Simple language.
- Help understanding what's happening behind a loading animation/page.



Why - The purpose

To explain:

- To explain a hidden digital process in a way that feels accessible.
- To reveal the complexity of something that people might think is “just waiting”
- To transform code logic into a graphic narrative.
- To make digital configurations visible and explain them to the audience.



What - Content focus:

Making visible what users never see:

- Code execution
- Server communication
- Asset loading
- Optimisation steps
- The actual complexity of “waiting”

Guiding question:

How do loading screens work?

What - Content focus:

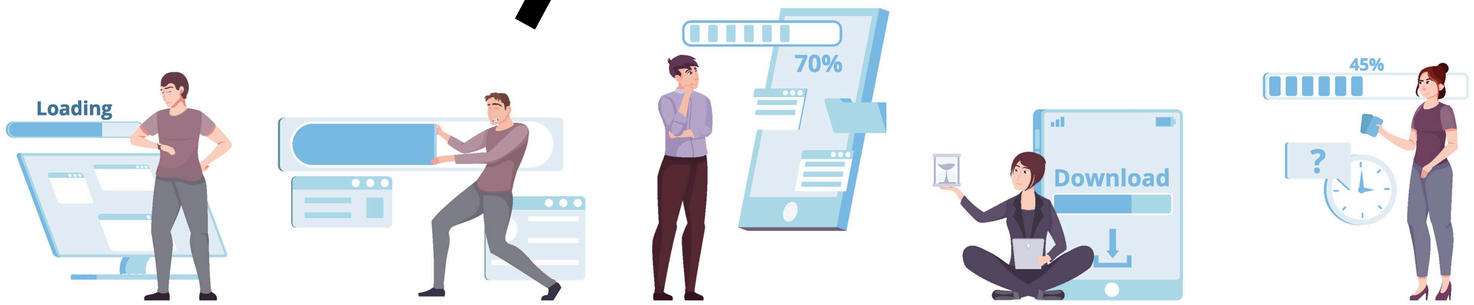
Sub questions:

- What triggers the loading screen indicator?
- How do different systems show and manage loading (old bars, spinners, shimmer effects, Lottie)?

Technical concepts to explore:

- Rendering
- Server responses
- Asset downloading
- Async JavaScript

Part 2;



Understanding the “Invisible”

The loading screen

A simple interface hiding complex process:

A loading indicator is an animated spinning or dot icon that shows the users that the content is being loaded. This helps to reduce user anxiety and reassures the user that the screen is executing a loading process and is not frozen.

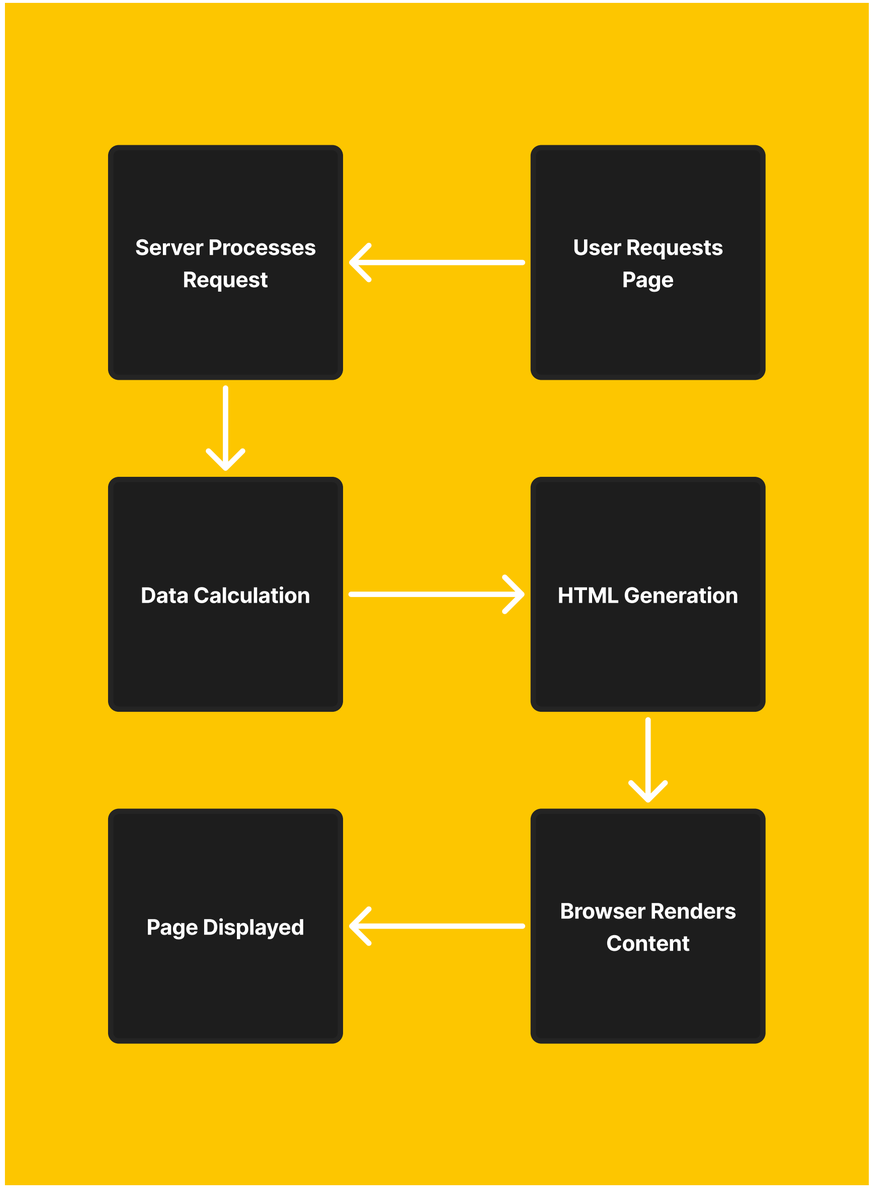


Diagram made by me

Triggering & UI

What triggers the loadig screen indicator:

Loading pages appear when a user does something that requires the system a moment to process information to “think”, meaning to perform calculations or render content. This can happen when navigating to another page, submitting a form, or carrying out an action that needs extra time for the system to calculate and load data before it is ready.

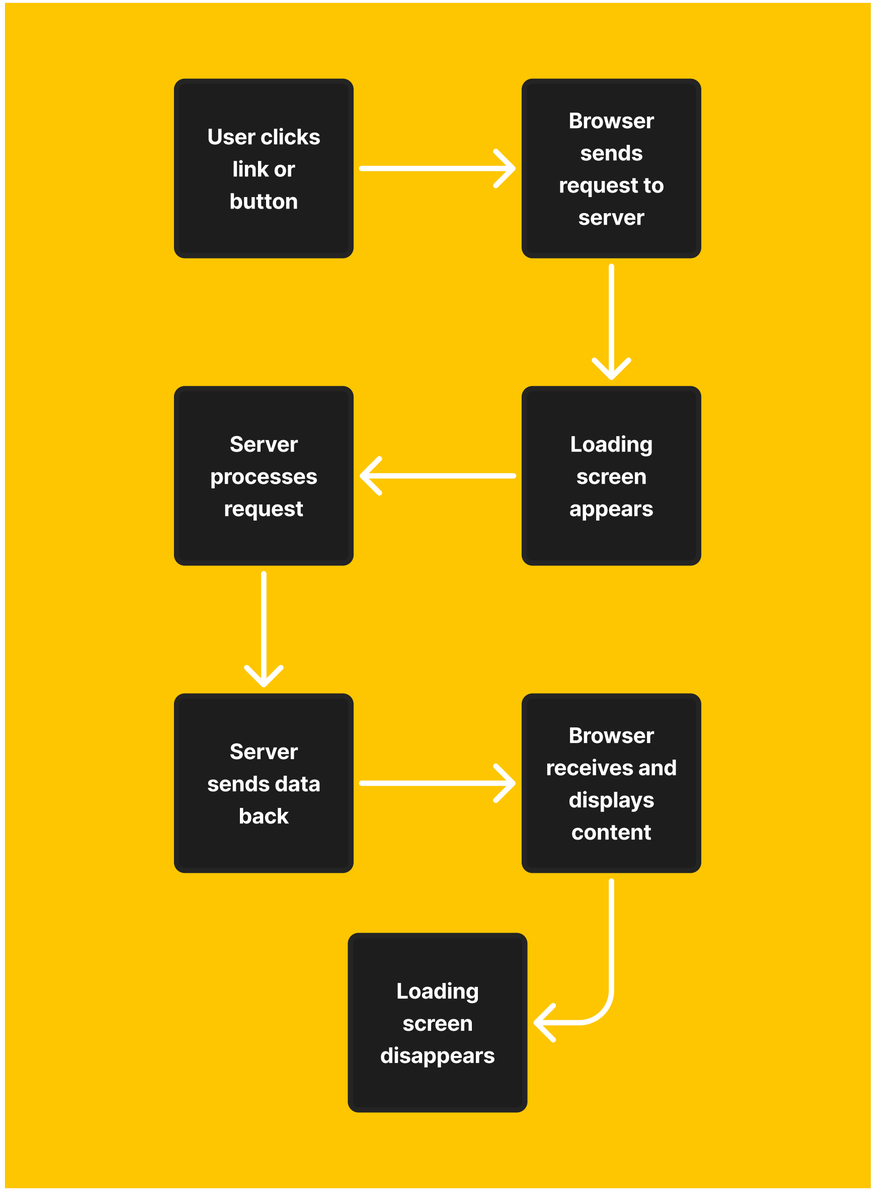
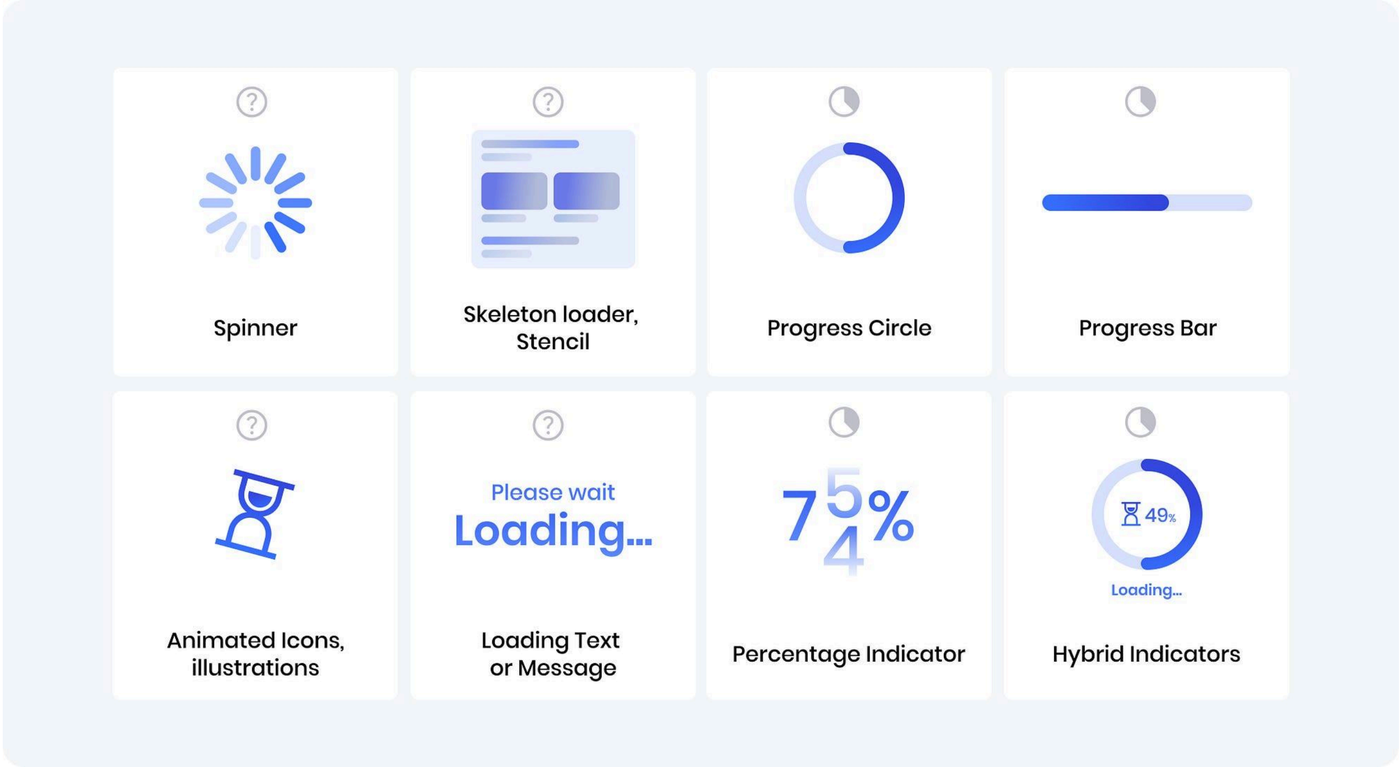


Diagram made by me

System Processes

How do different systems show and manage loading:

UI patterns like progress bars, spinners, skeleton screens, and animated Lottie effects provide feedback on system activity. They help users understand that work is being done in the background, reducing perceived wait time and frustration.



What happens in the background

The invisible process behind it:

When a page loads, the browser opens HTML, CSS and JavaScript to make the DOM (document object model), meanwhile, the server request and obtain resources like images, fonts, and scripts.
The final step is painting the pixels to the screen.

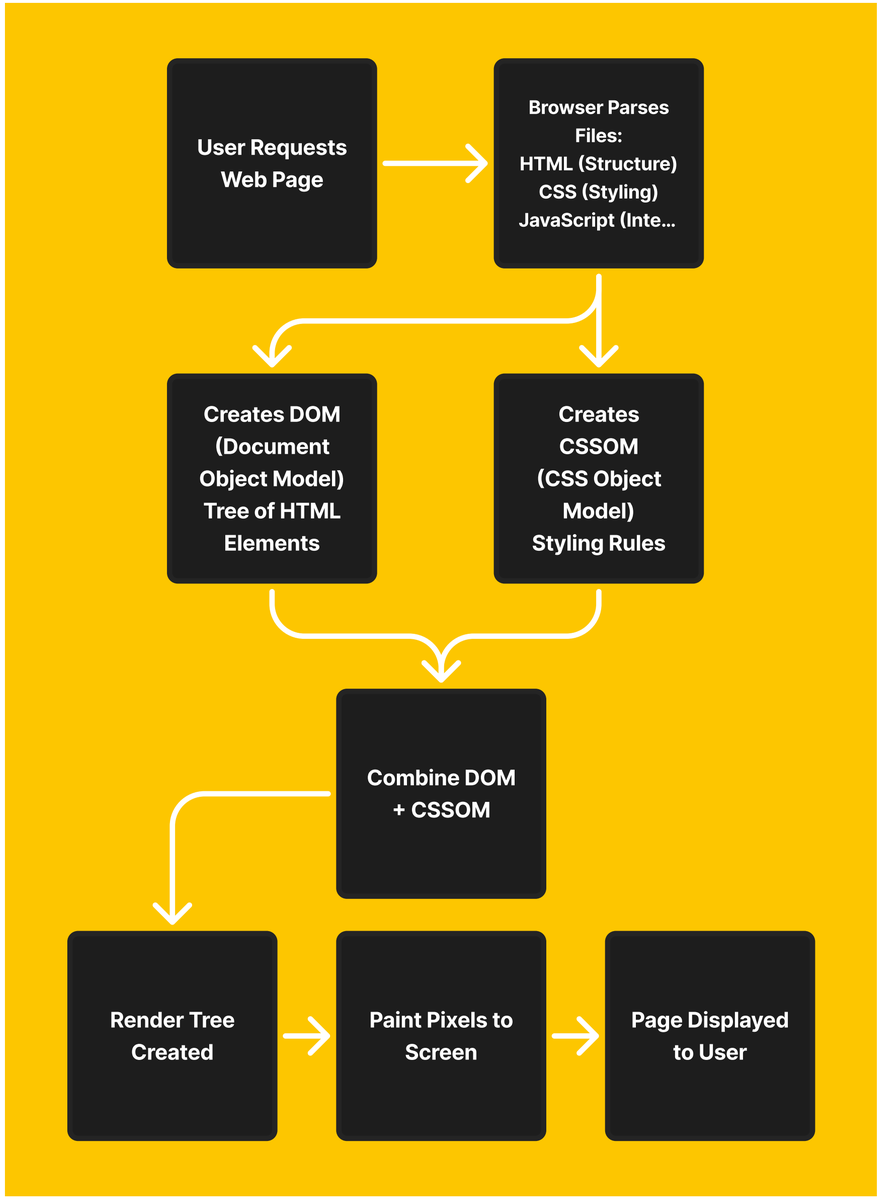
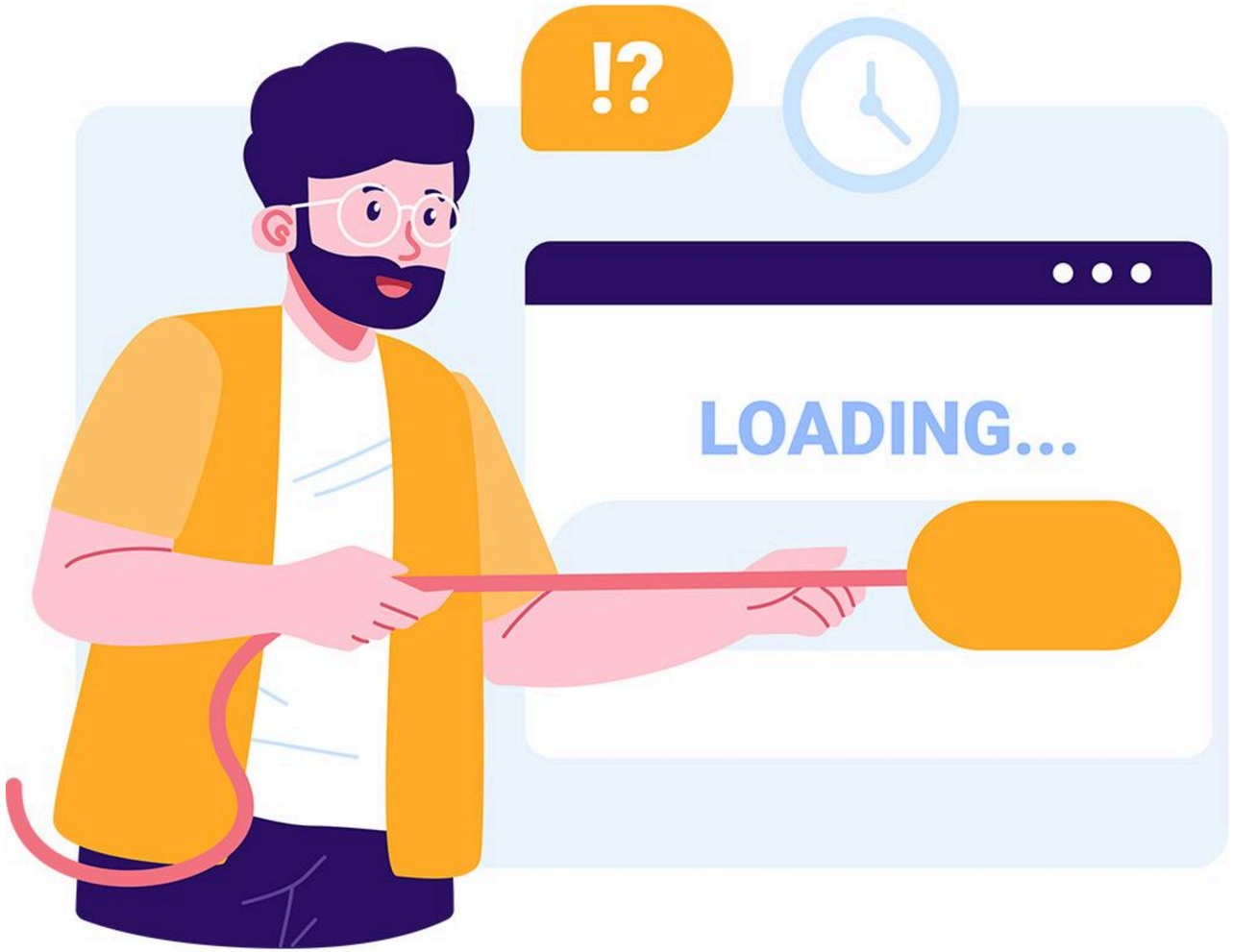


Diagram made by me

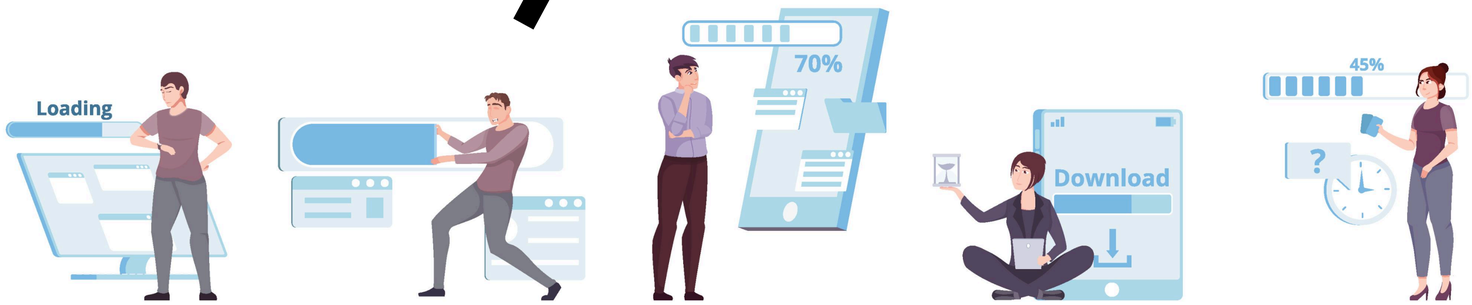
User perception

Why does this matter to the user?:

Users perceive waiting time differently depending on the feedback provided. By making the loading process visible, users understand that the system is actively working. This process improves patience, trust, and perceived performance.



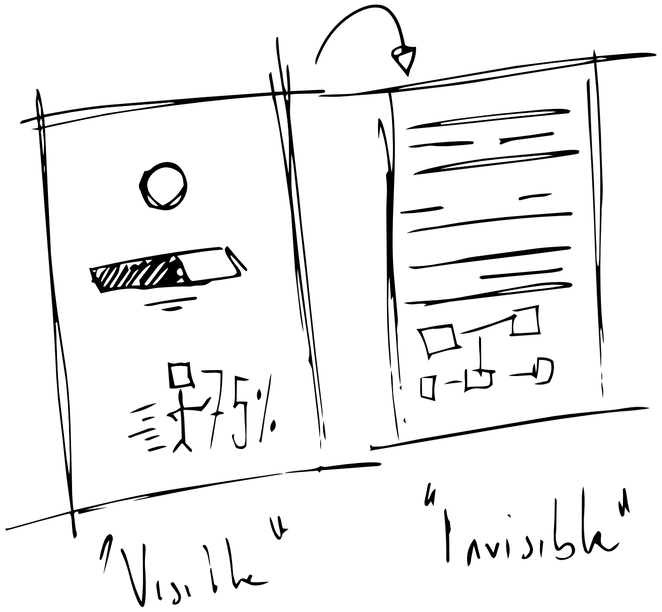
Part 3;

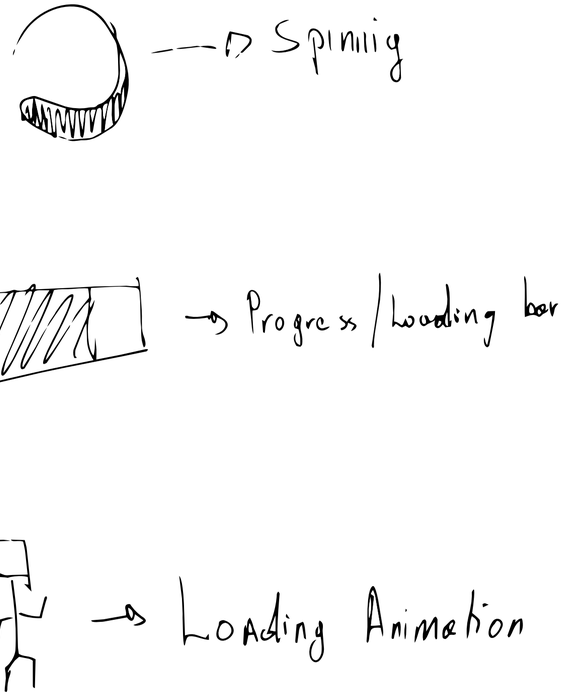


First ideas and concepts

Design Concept / Ideation

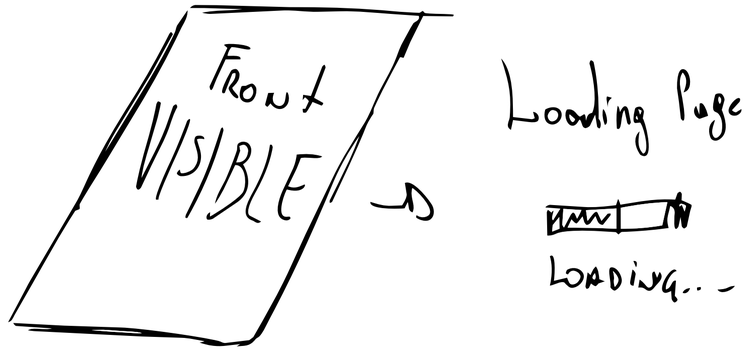
Initial idea:
My initial idea was to create a double-sided poster. The front side would show a loading screen, while the reverse side would reveal the code and diagrams illustrating the system processes and calculations that trigger and support the loading page.

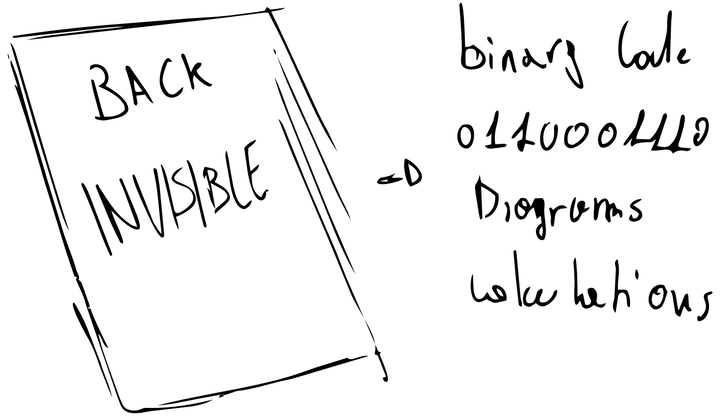


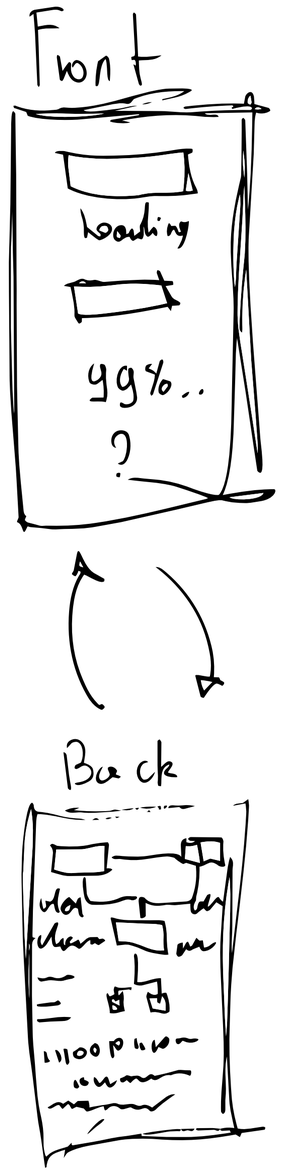


Design Concept / Ideation

Limitations of the Approach:
The double-sided poster required users to flip between two sides, which separated information in space rather than helping them understand how processes coexist during loading.

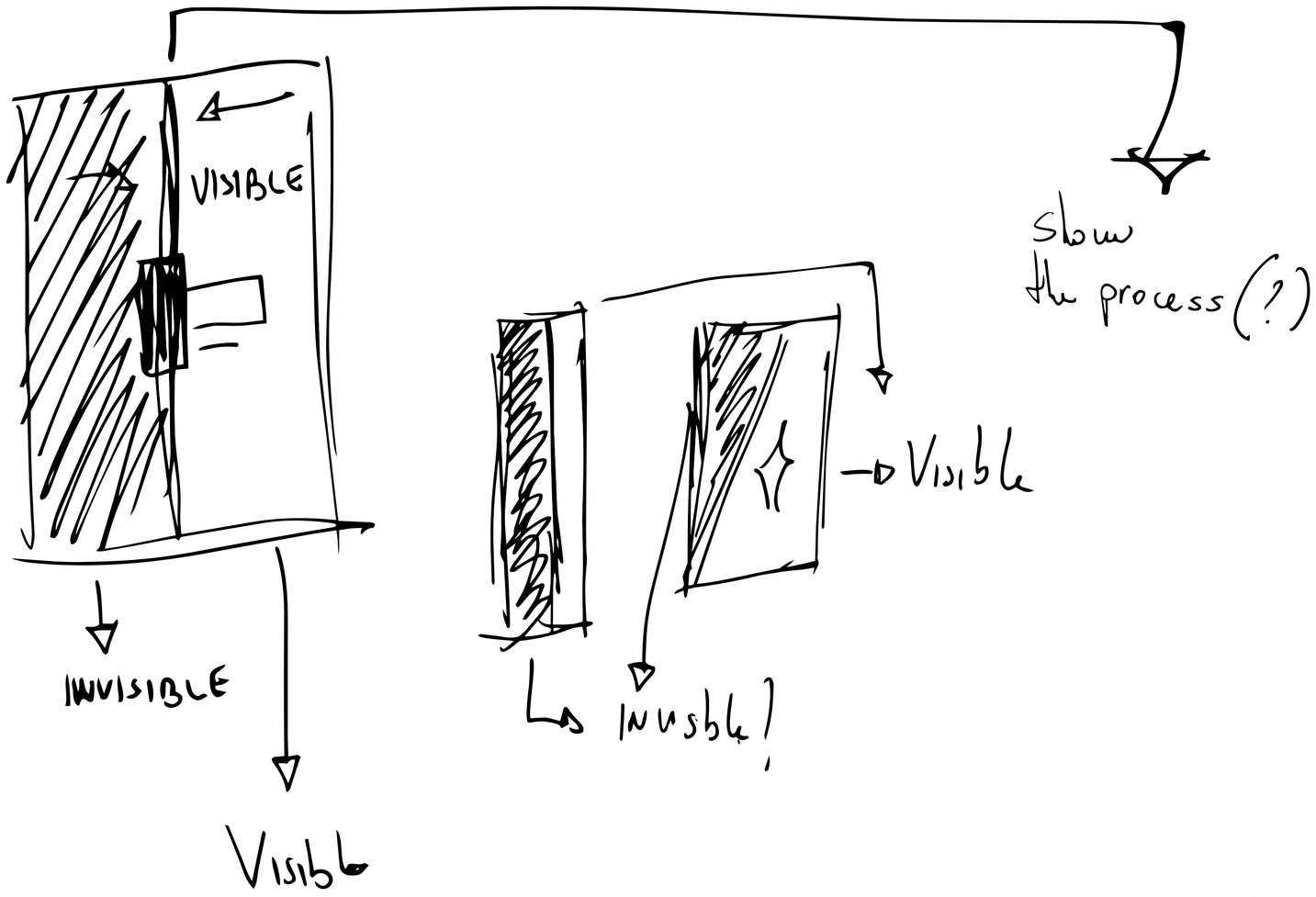






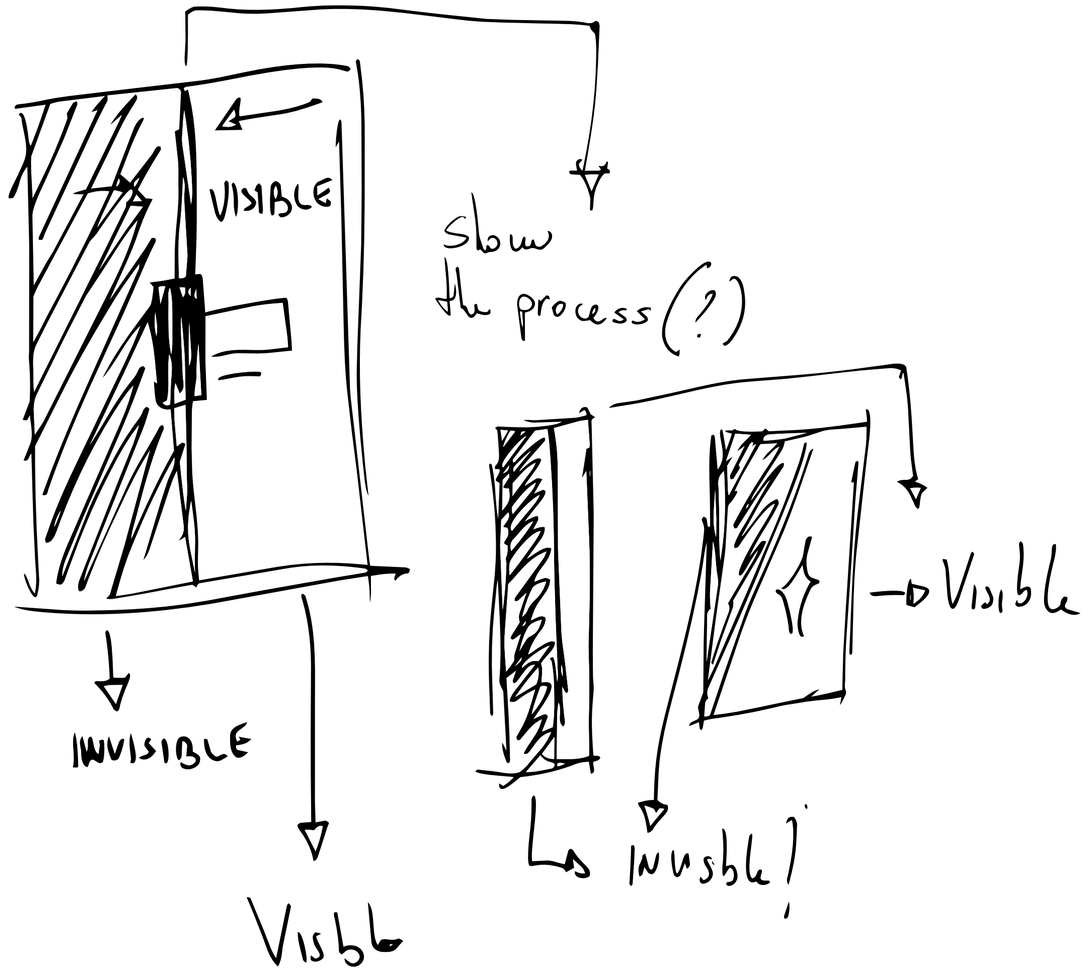
Design Concept / Ideation

Second idea:
After exploring a double-sided format, I considered a split poster layout as an alternative approach. This idea involved dividing the poster in the middle, with the visible loading screen on one side and the invisible system processes on the other.



Design Concept / Ideation

Limits of Visual Separation:
While this approach created a clear visual distinction between the two layers, it framed them as separate states rather than communicating how they operate simultaneously during the loading process.



Design Concept / Ideation

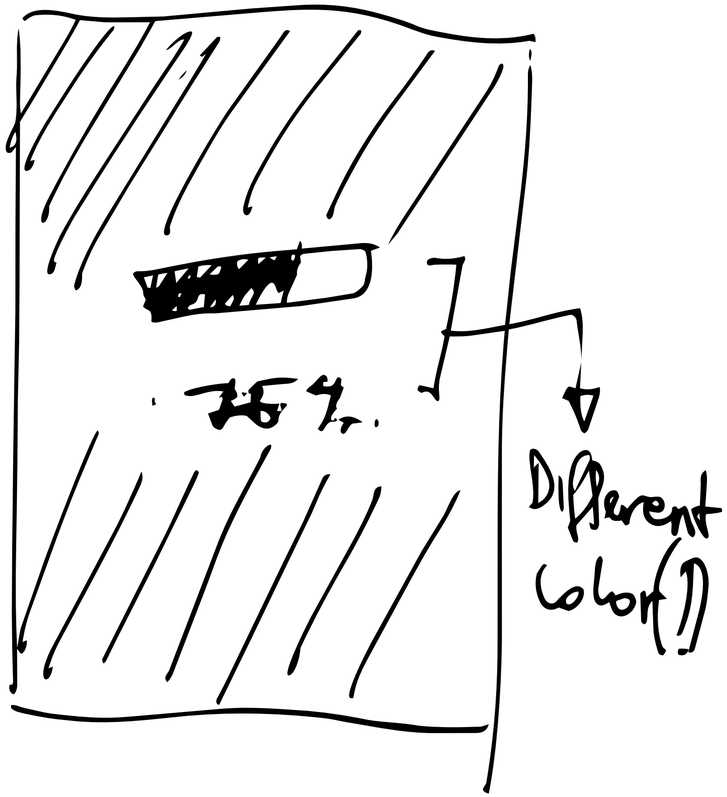
Chosen Visual Strategy:

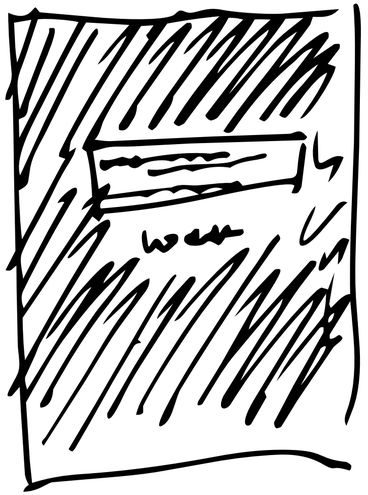
I started to consider an approach that would allow the visible and the invisible elements to coexist within the same space, without separating the two elements in “strict” way



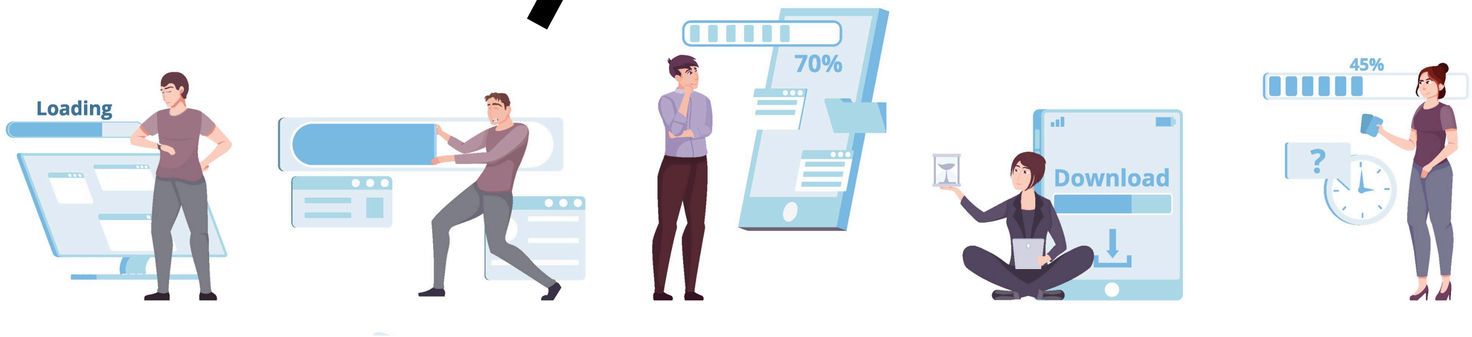
Design Concept / Ideation

Chosen Visual Strategy:
After researching into techniques that allow layered information to remain distinct while occupying the same space, i came across anaglyph design





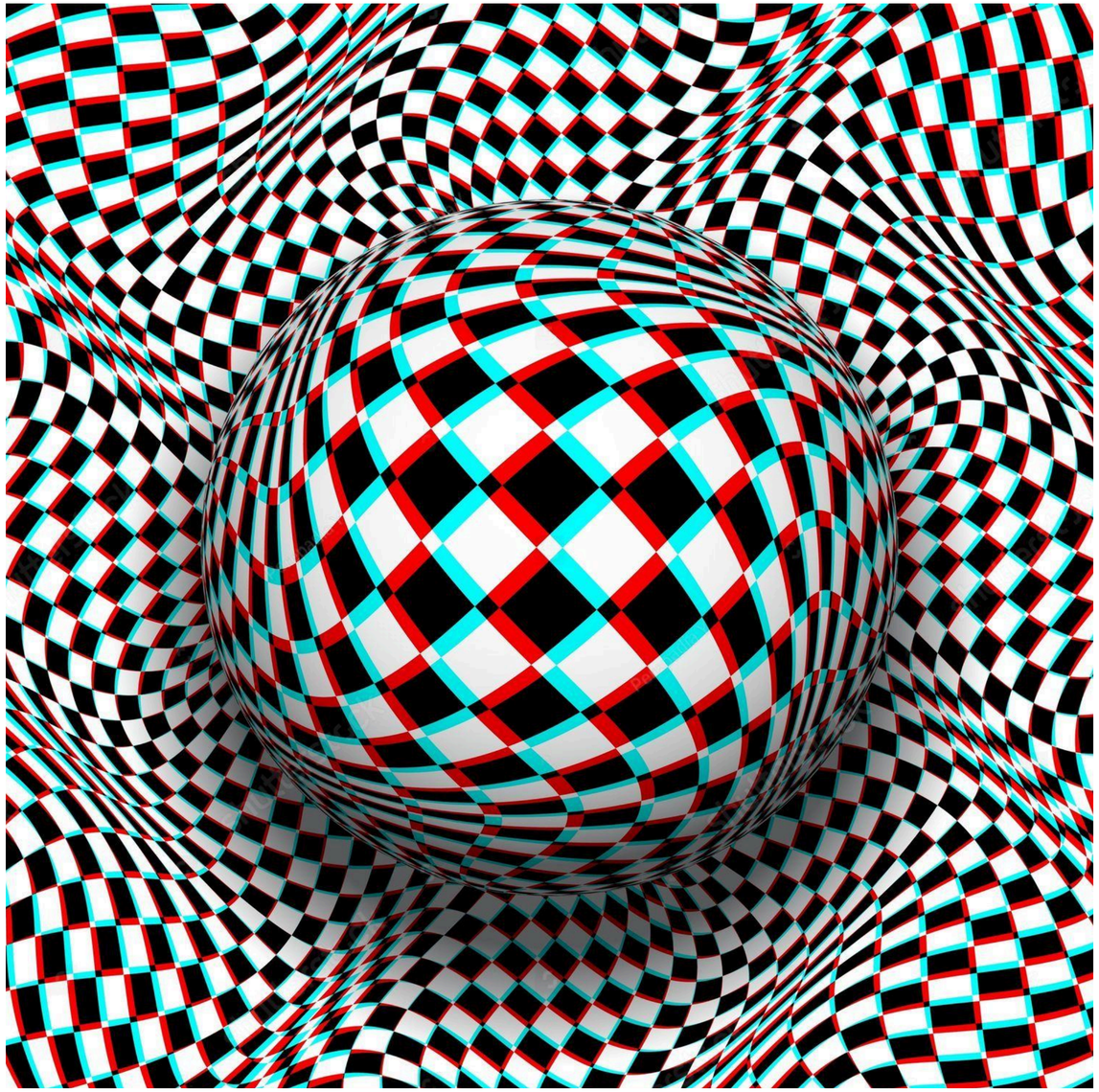
Part 4;



The final idea

Anaglyph Design

what is an anaglyph and how it works:
An anaglyph is a type of stereoscopic image, meaning that while the image appears three dimensional when viewed, it is actually two-dimensional. This characteristic allows both the visible and invisible elements to occupy the exact same space while remaining visually distinct.



Anaglyph Design

How to separate the two elements:
At first, the image may appear visually complex, due to the overlapping red and blue elements. However, there is a way to selectively “hide” one layer in order to reveal the other. This is achieved through the use of special glasses, featuring one blue lens and one red lens.

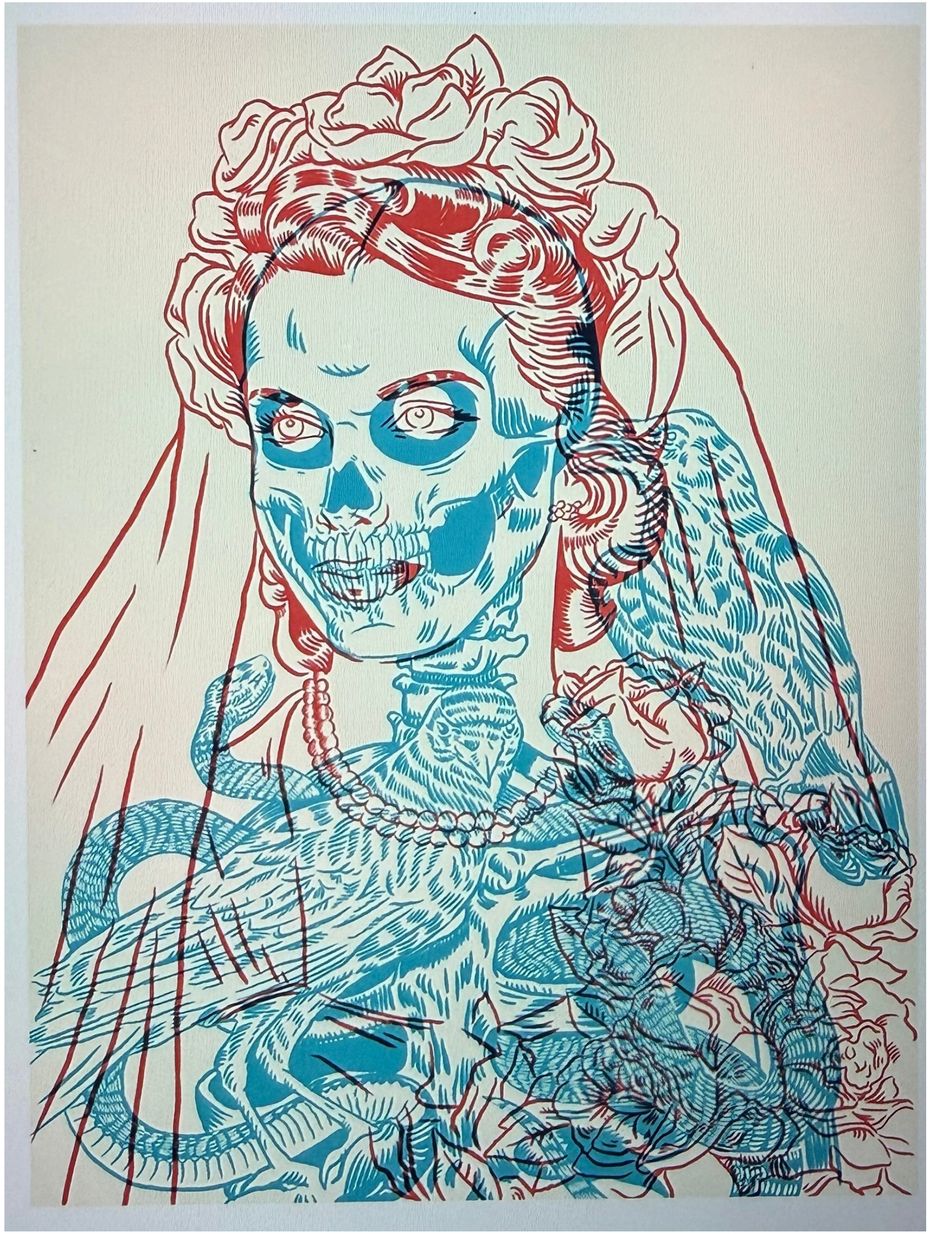
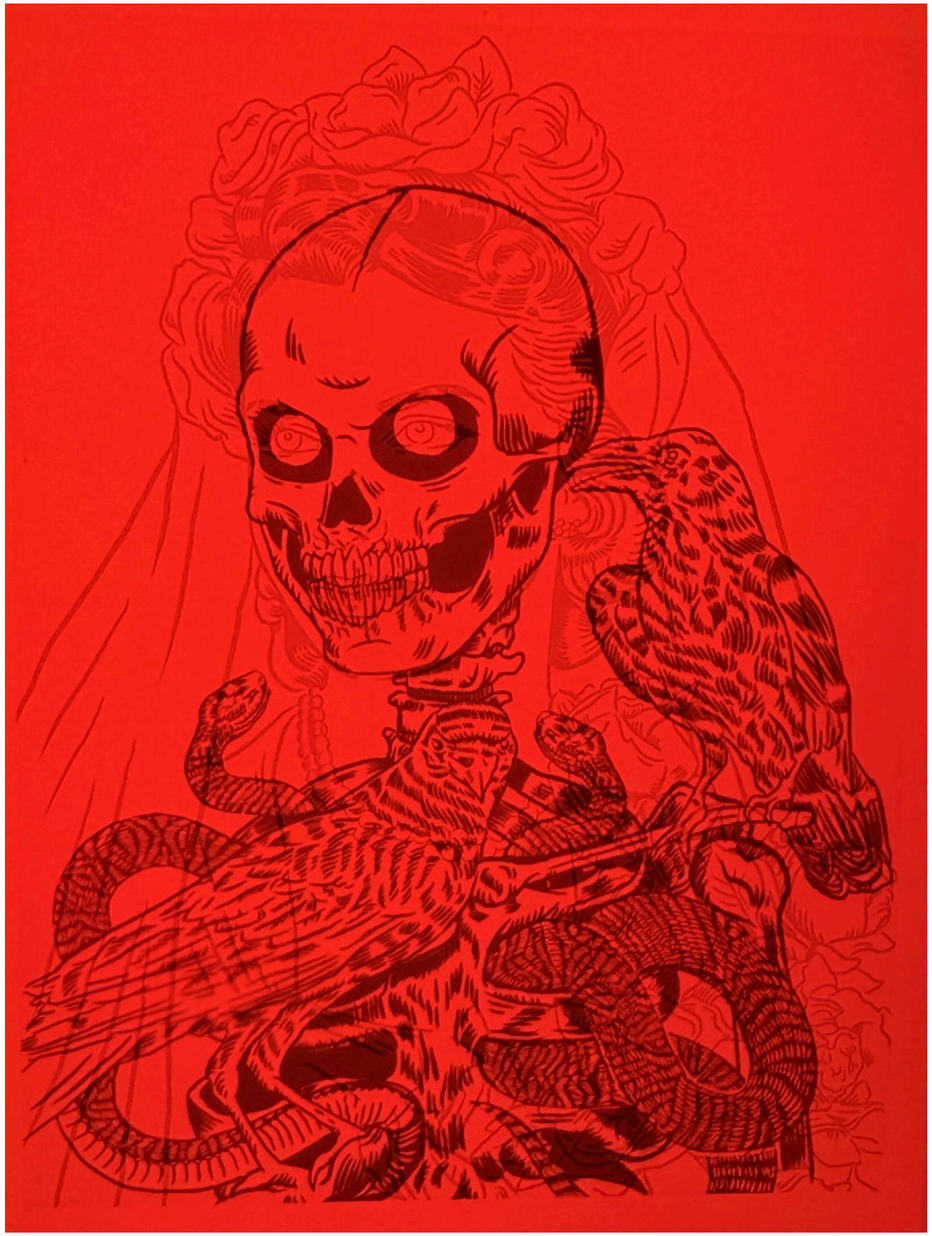


Image credit: Travis Knight

Anaglyph design viewed through the Blue lens:



Anaglyph design viewed through the Red lens:



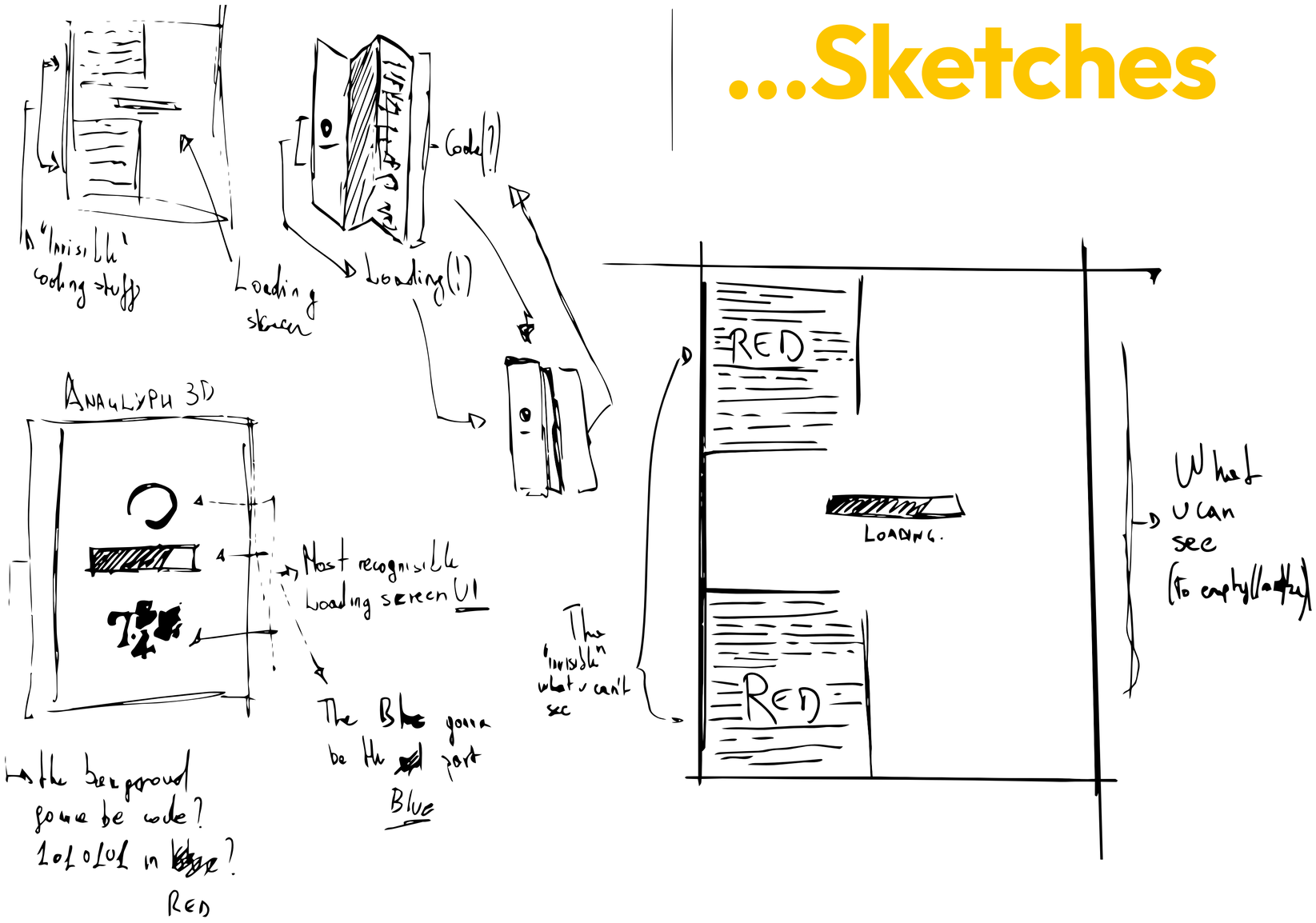
Additional...

...Sketches

With anaglyph design as the chosen direction, I began exploring multiple ideas that could support this approach.

Initially, I continued to think in terms of separating the visible and invisible into distinct sections, using clear lines to divide the two. I also explored ideas such as brochure-like formats, which ultimately did not align with the anaglyph approach.

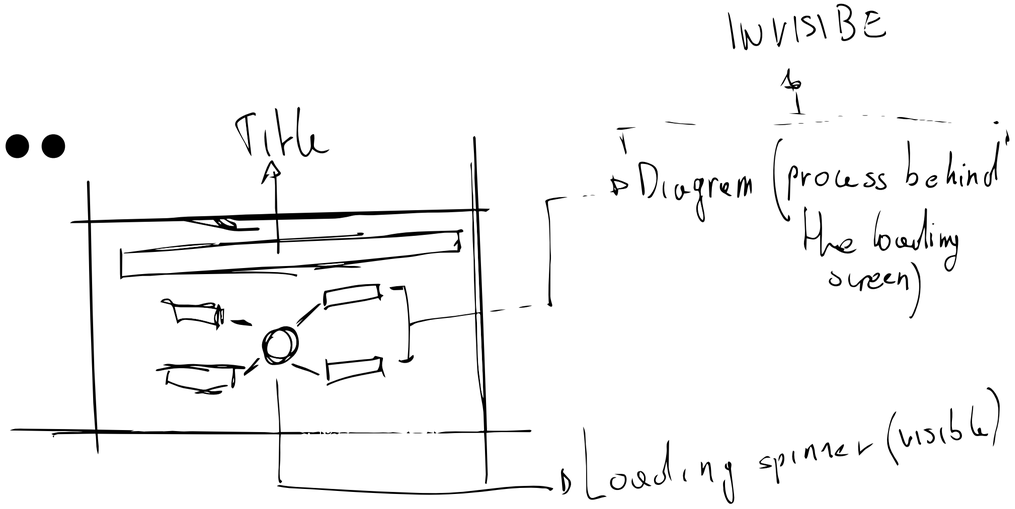
I then began to shift my focus towards the idea of “blending” the two elements, rather than keeping them strictly separated.

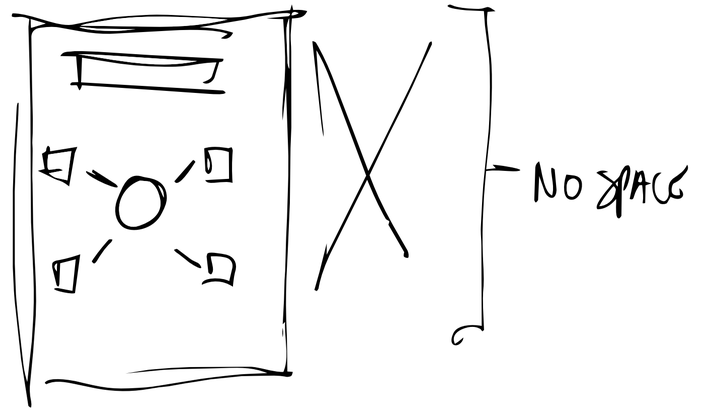


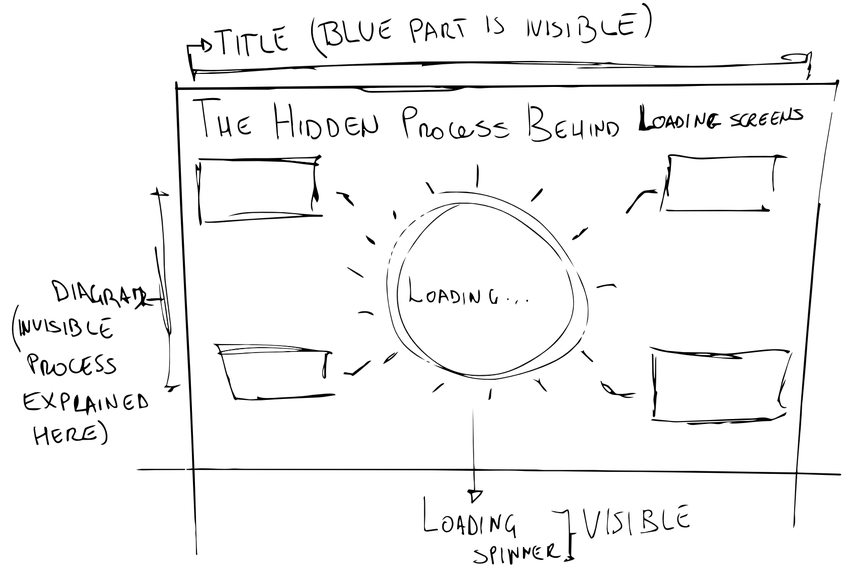
Additional...

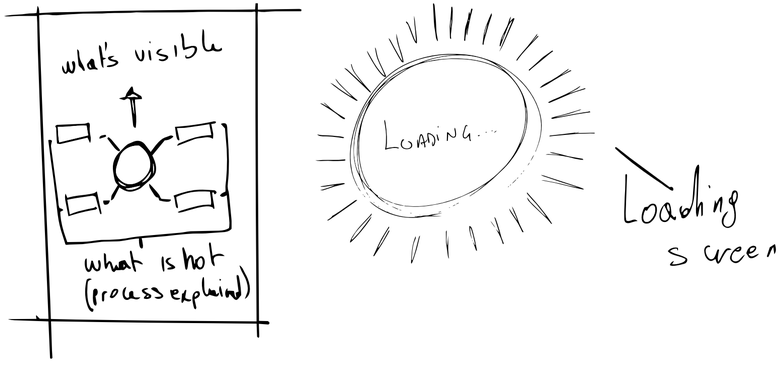
...Sketches

As I continued sketching and developing ideas, I realised that a portrait format was not the most effective in terms of space optimisation. I therefore decided to explore a landscape orientation, which allowed for a clearer structure and provided more space to develop the visual narrative.



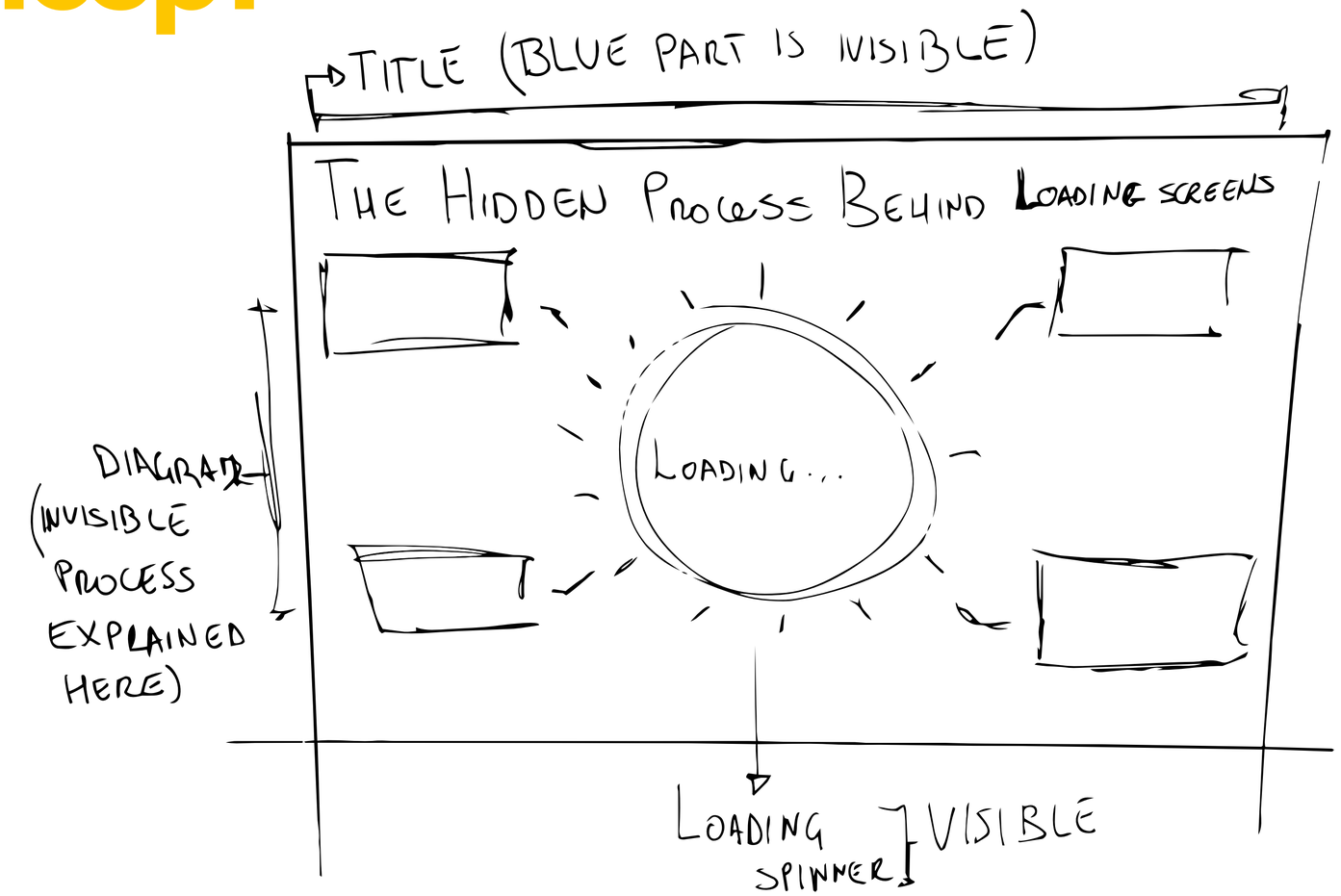




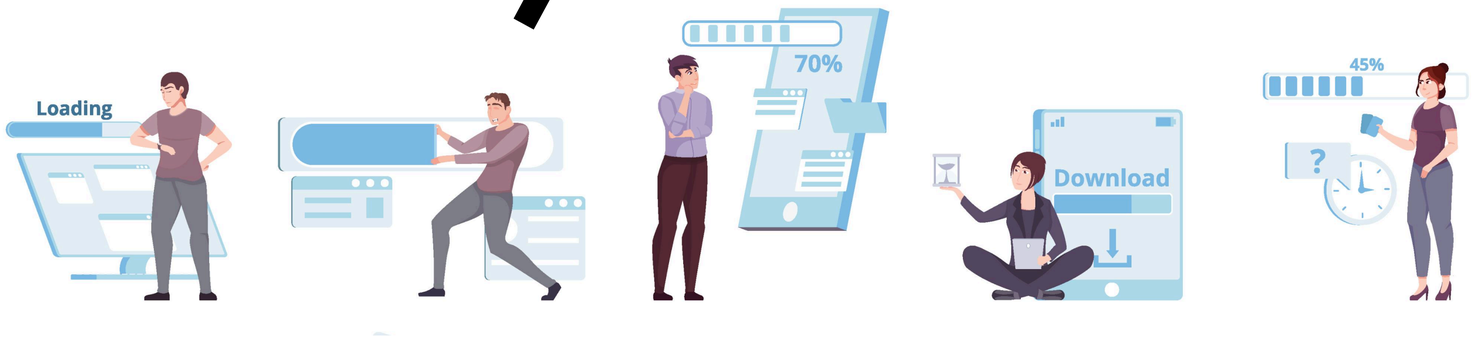


Final Design Concept

The idea:
The final idea emerged while revisiting the sketches developed in a landscape orientation. I liked the idea of creating a horizontal poster featuring an “invisible” diagram that explains all the stages occurring behind a loading screen, with a “visible” spinner placed at the centre to represent the page loading process.



Part 5;



Development phase

The colours

Through testing different anaglyph designs using 3D glasses, I observed that the visibility of colours varied depending on the colour combinations used. In some cases, the blue elements appeared more prominent when viewed through the red lens, and the red elements through the blue lens. In other tests, the effect behaved as expected, with blue elements visible through the blue lens and red elements through the red lens.

Image credit: Creative Market on pinterest



image without using 3d anaglyphic glasses



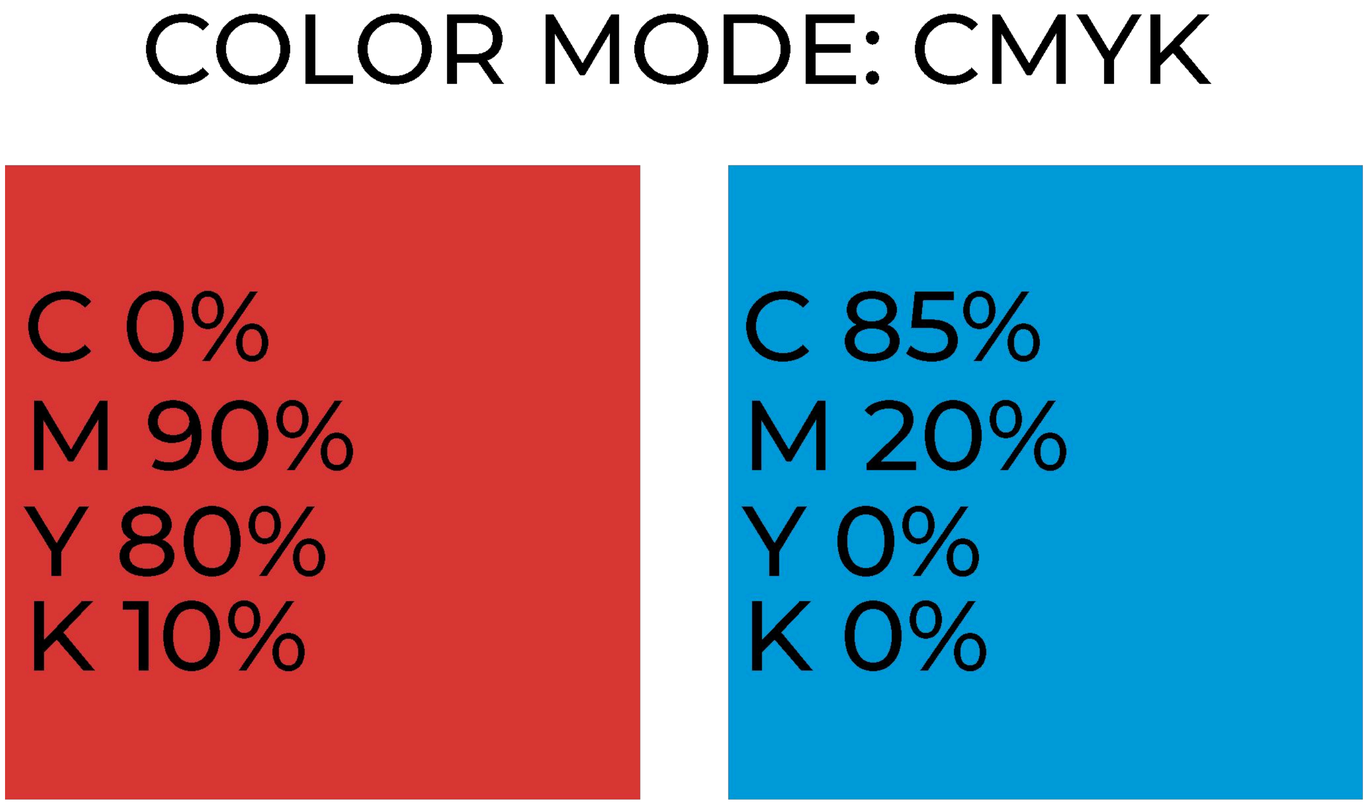
image using the red lens of the3d anaglyphic glasses



image using the blue lens of the3d anaglyphic glasses

The colours

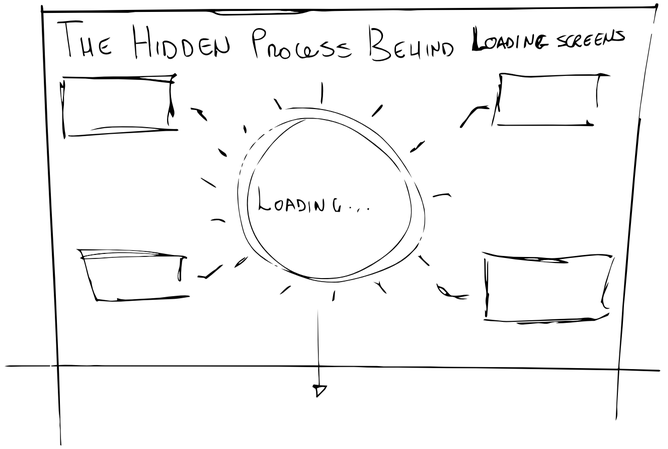
Further research revealed that this behaviour depends largely on the colour values used. As the final outcome was intended for print, I worked in CMYK (Cyan, Magenta, Yellow, and Key, which is black) colour mode within Illustrator. After discussing this with a tutor in the LCC digital print studio, I selected specific CMYK colour percentages to achieve maximum colour saturation and improve the optical effect of the anaglyph.

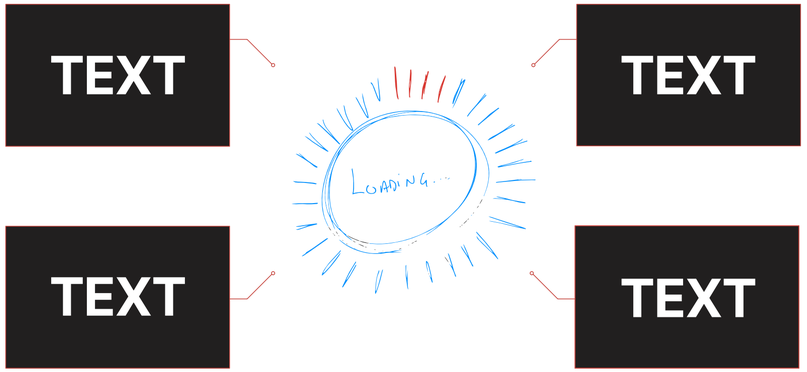


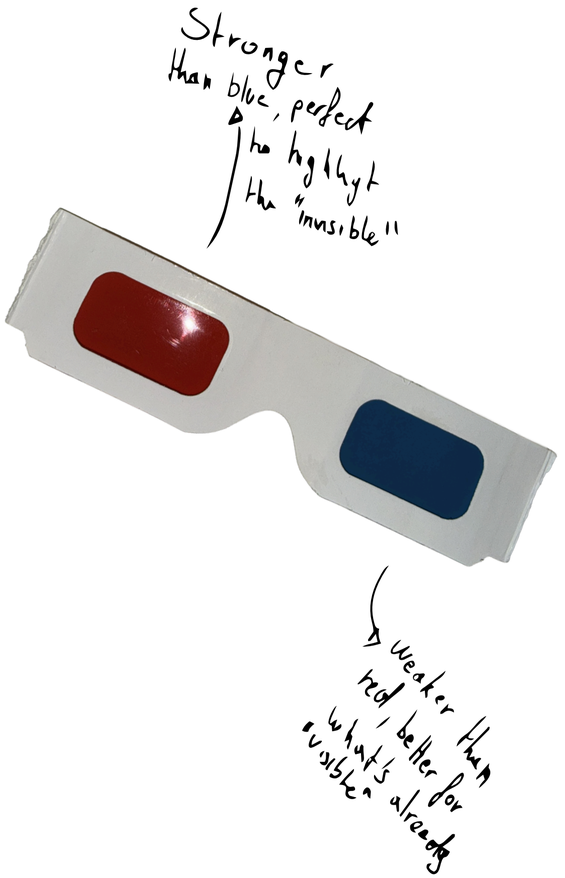
Made by me

The project

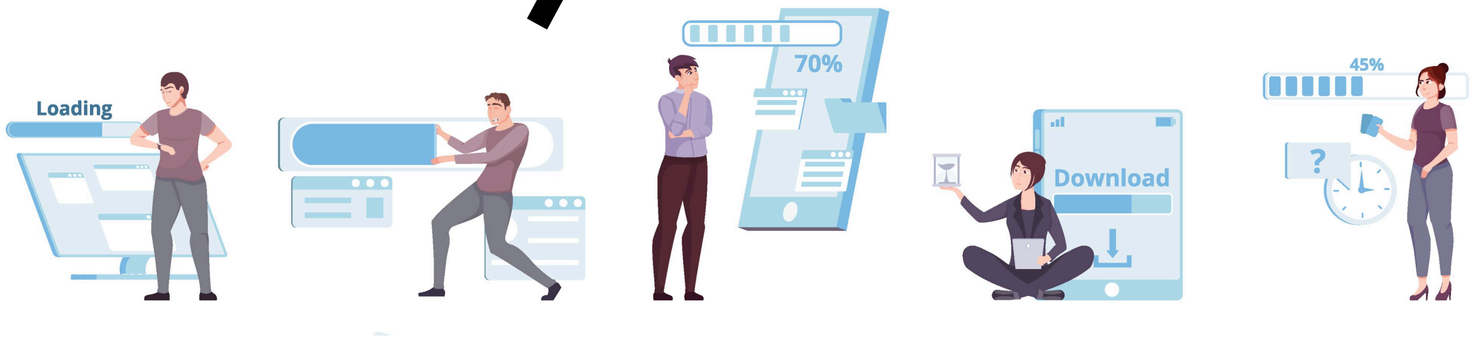
I decided to represent the invisible elements using red, and the visible elements using blue. Through testing different anaglyph designs with 3D glasses, I observed that red appears more prominent than blue. For this reason, I chose to use the more saturated colour to highlight what is normally hidden, while assigning the less dominant colour to elements that are already visible and familiar to users.





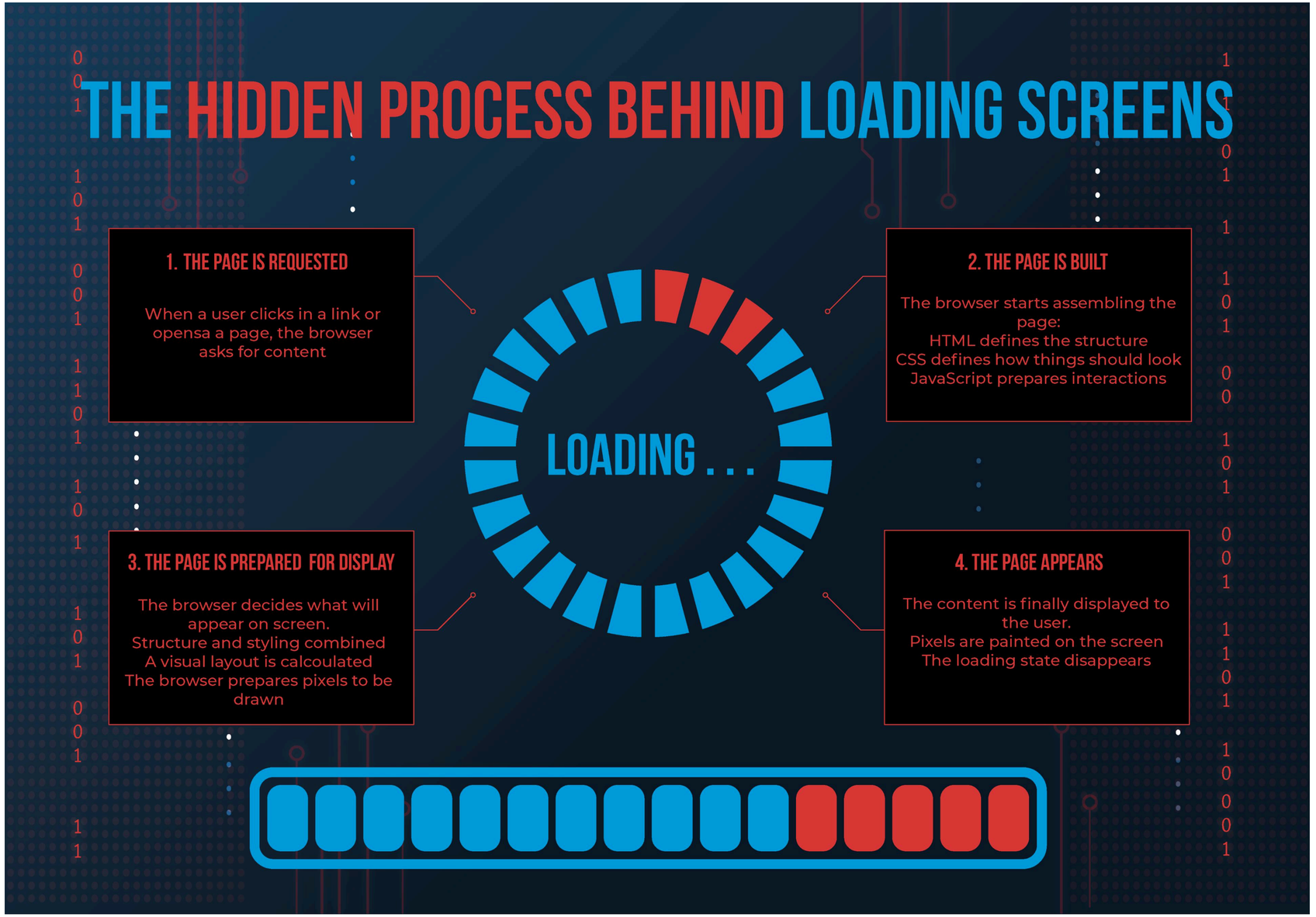


Part 4;

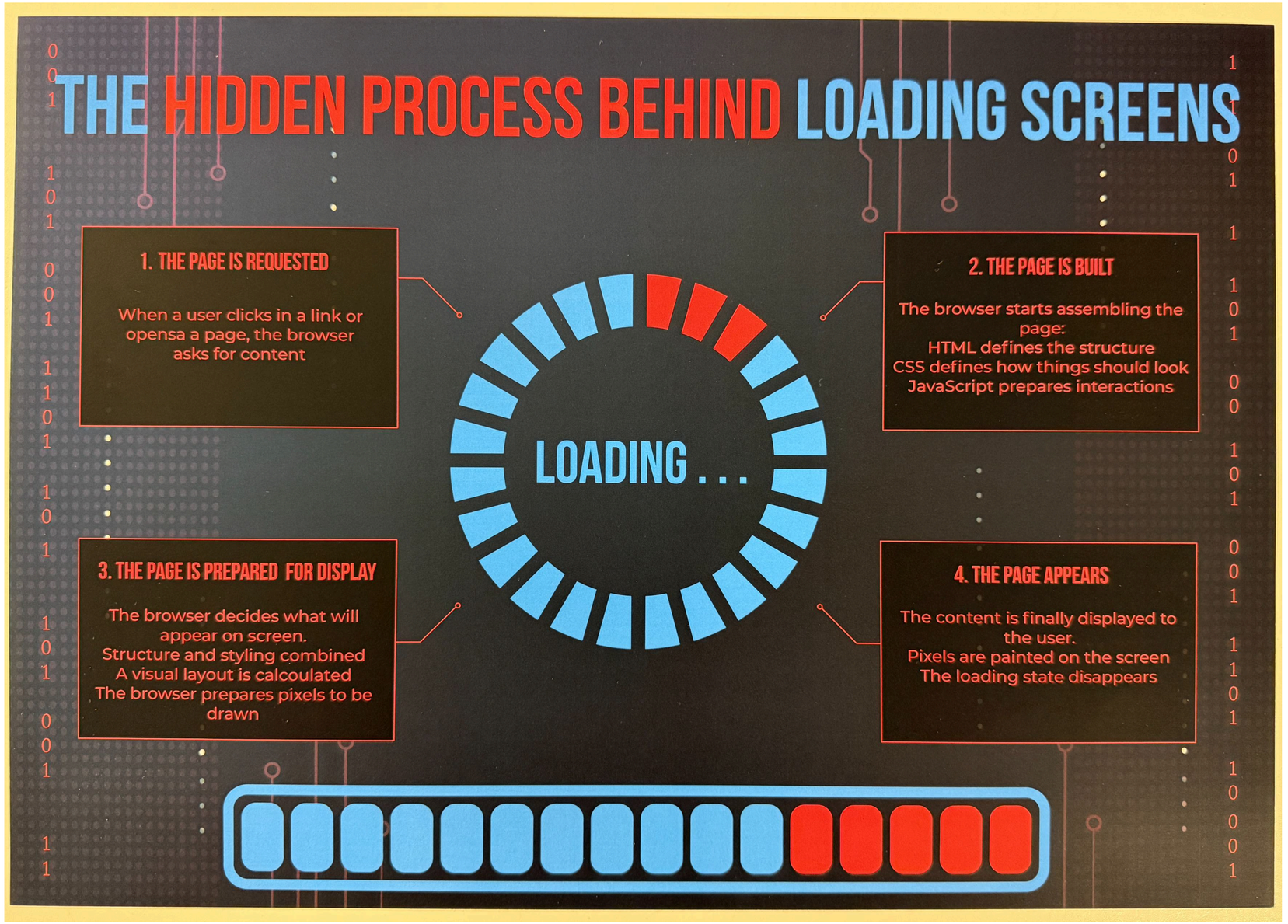


The project

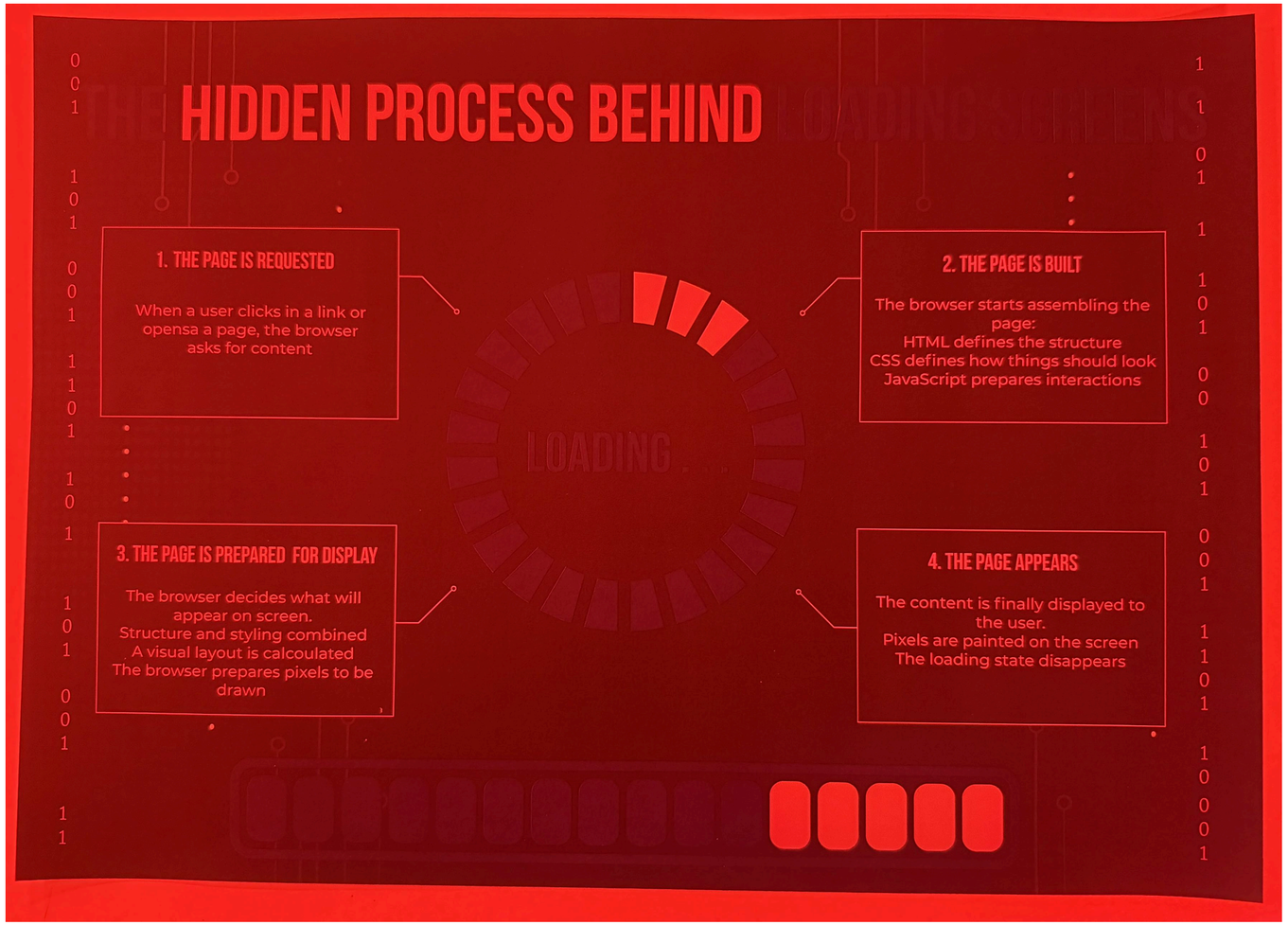
The Project (A3 Size, Digital)



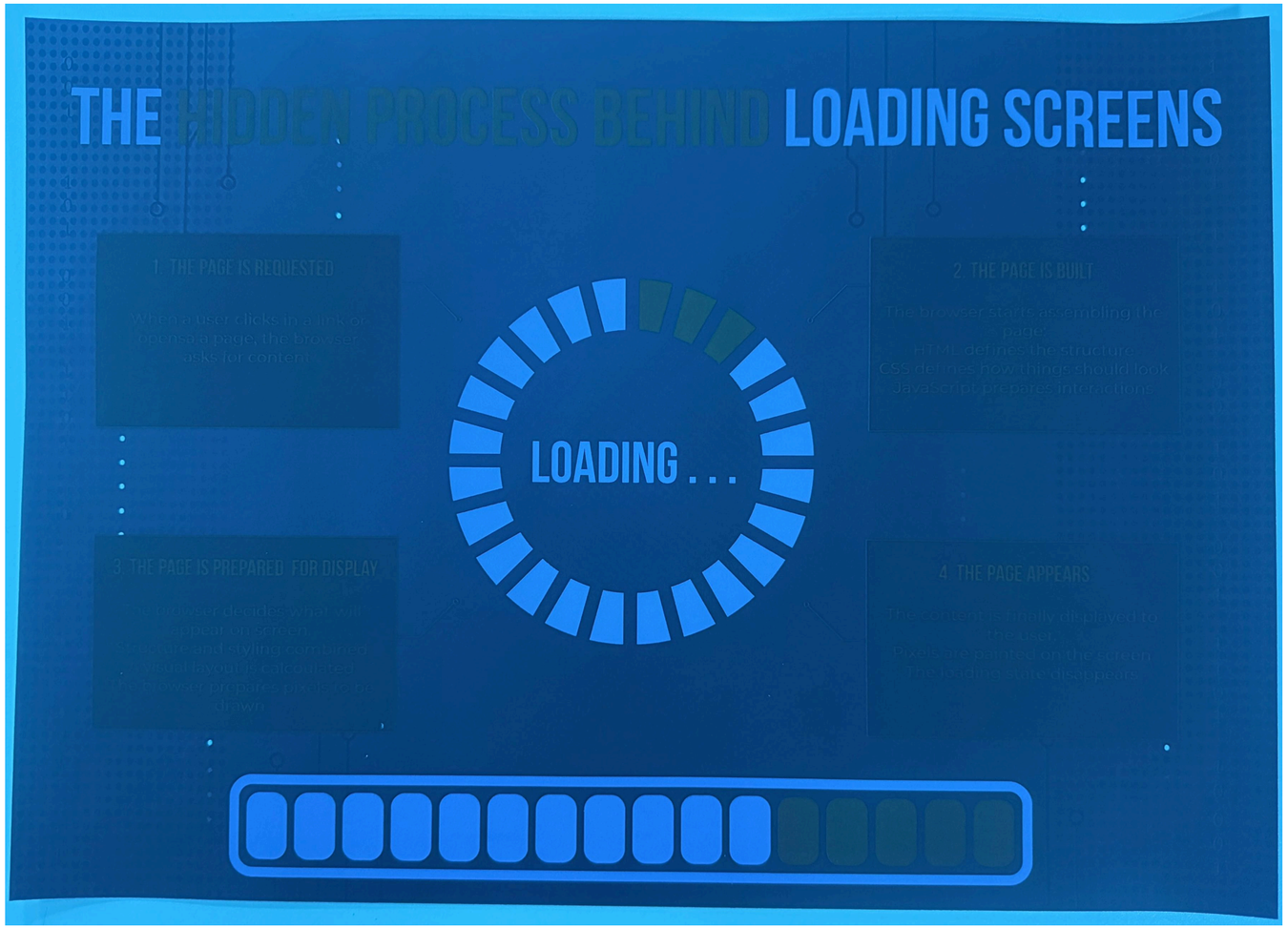
The Project (A3 Size, Printed)



The Project - Using the 3D Glasses (red)

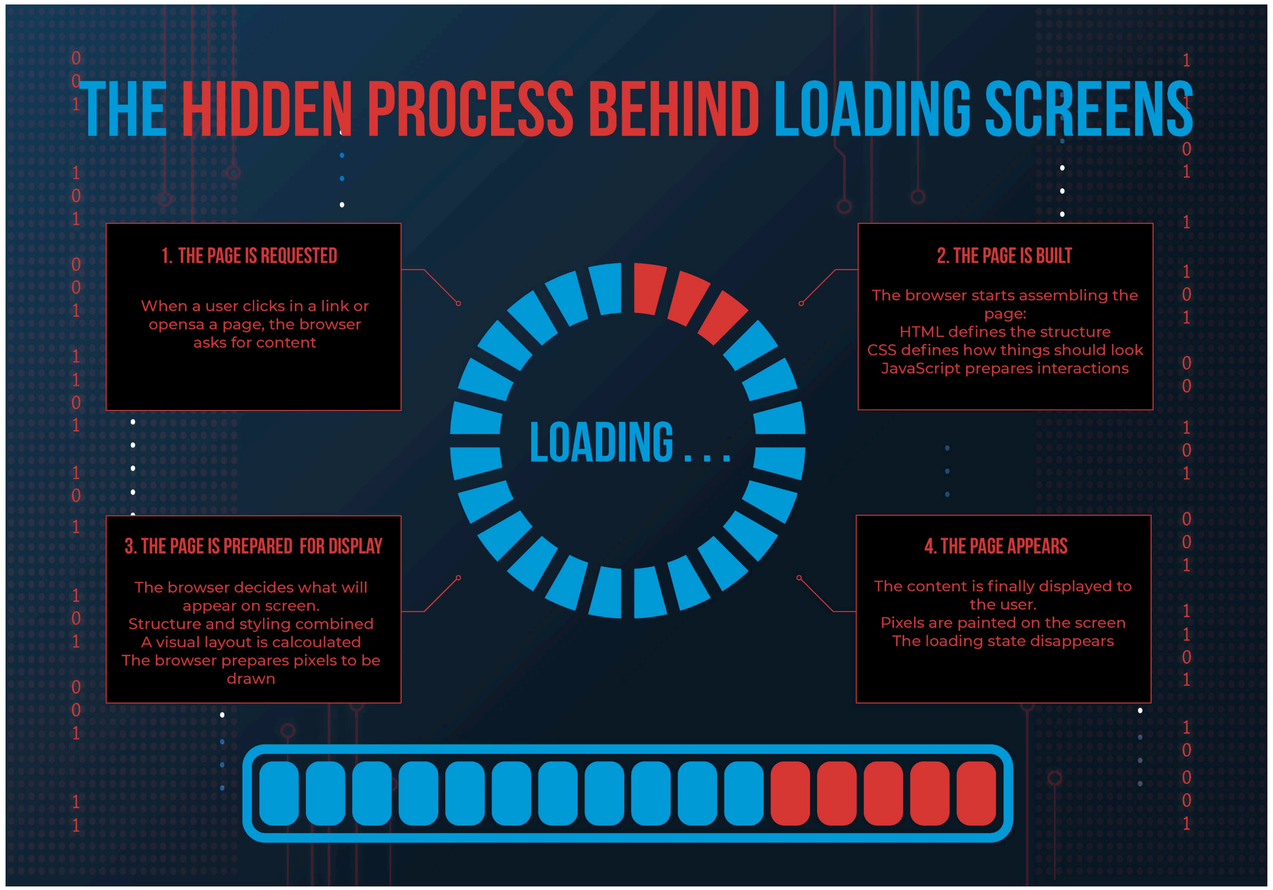


The Project – Using the 3D Glasses (Blue)



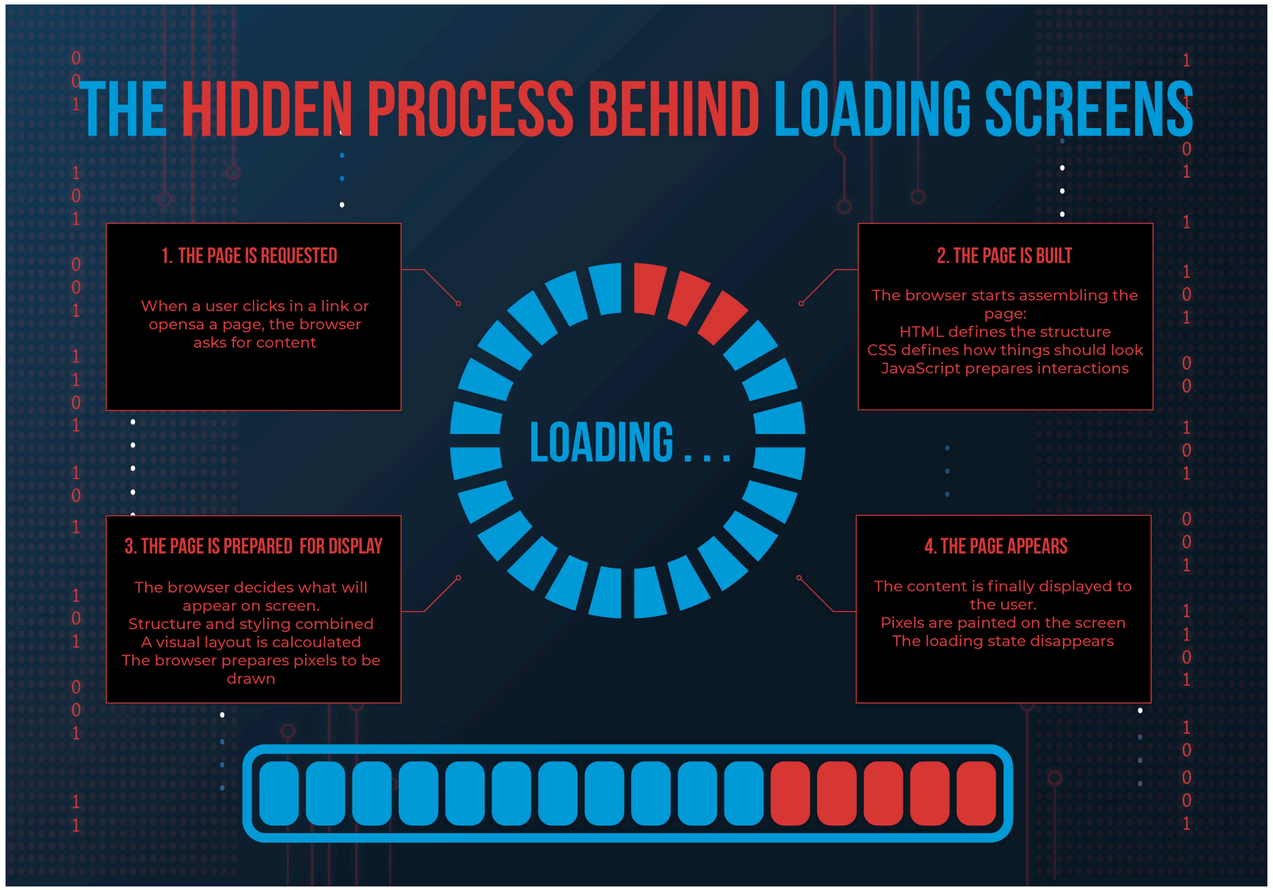
How this design works

The content represented in red corresponds to information that users do not normally see. This includes diagrams that explain the processes occurring behind a loading screen, which are displayed in red against a black background. When viewed through the red lens, this content becomes clearly legible, while it is effectively minimised when viewed through the blue lens.



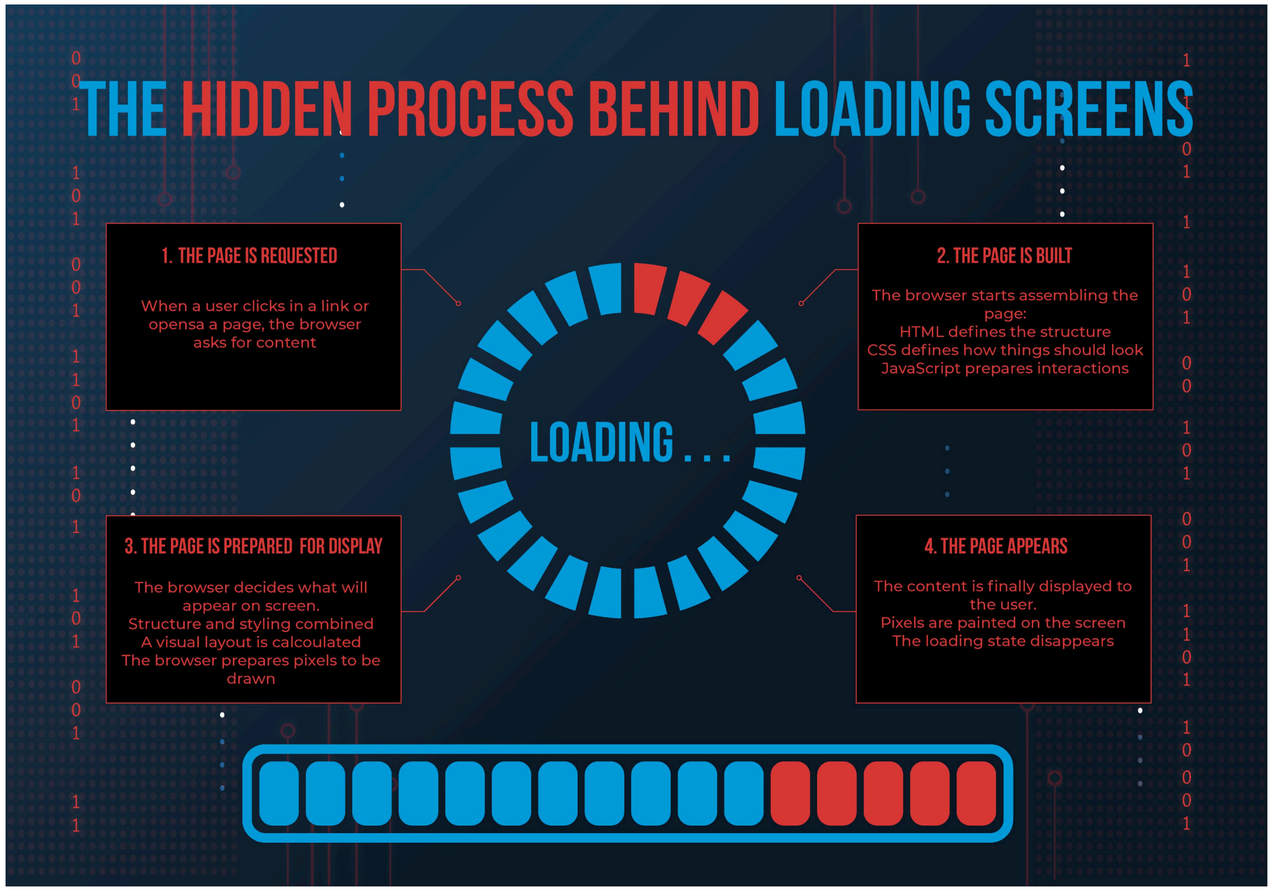
How this design works

The visible loading interface elements, specifically the loading spinner and the progress bar, are represented in blue. Selected elements within the progress bar are shown in red to indicate that the page has not yet fully loaded. This visual relationship allows users to understand that, while the loading interface is still active, background processes are taking place. By reading the diagrams, users gain insight into what is happening and what will occur once the page finishes loading.



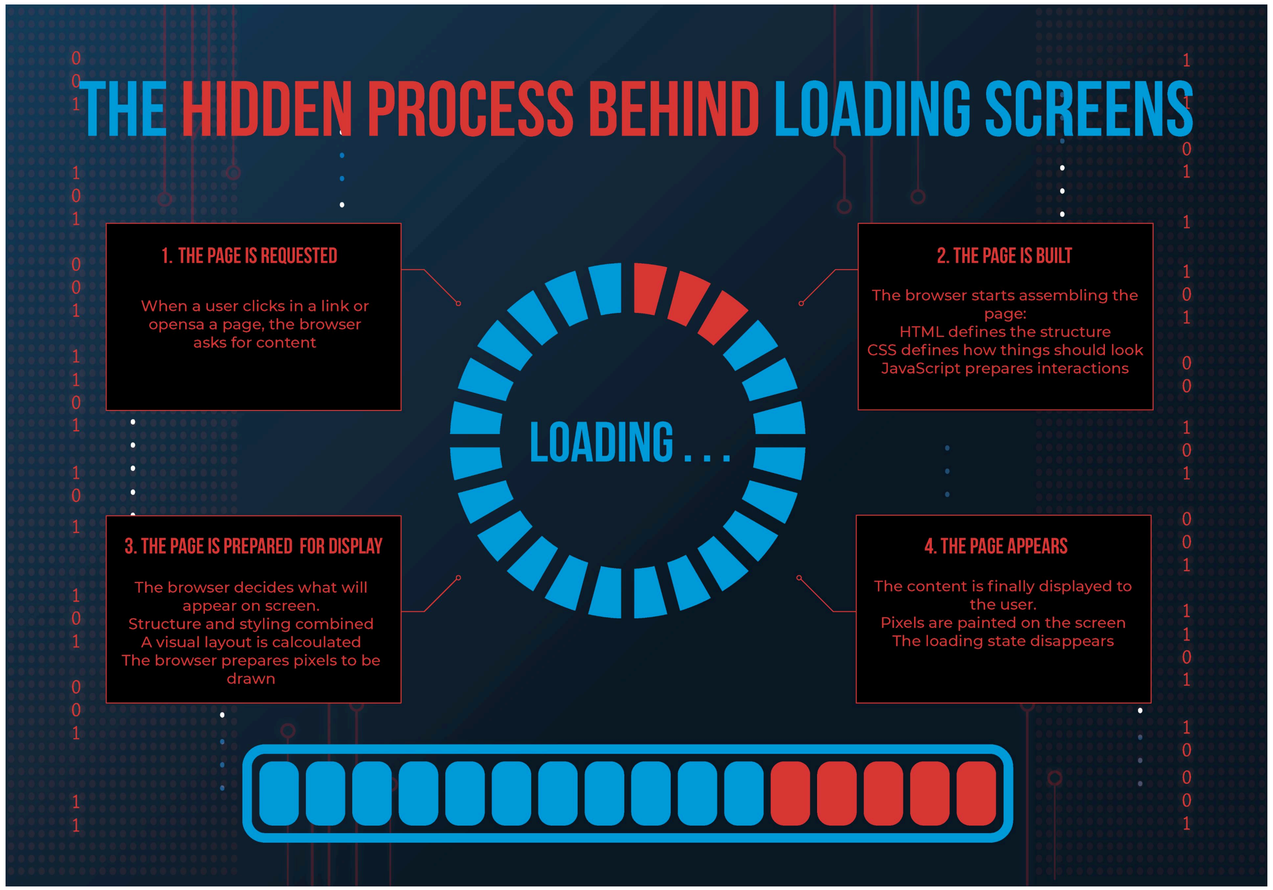
How this design works

Finally, I introduced some little details to enhance the overall visual cohesion of the poster. These include two binary code bars positioned on either side of the composition, representing the calculations performed by the browser during the loading process and some “circuit” graphic elements were added to further reference the digital nature of the topic. All of these supporting details were kept in red to remain consistent with the invisible layer of the system. The white and blue dots on the back ground are also little details I used to make the design more particular.

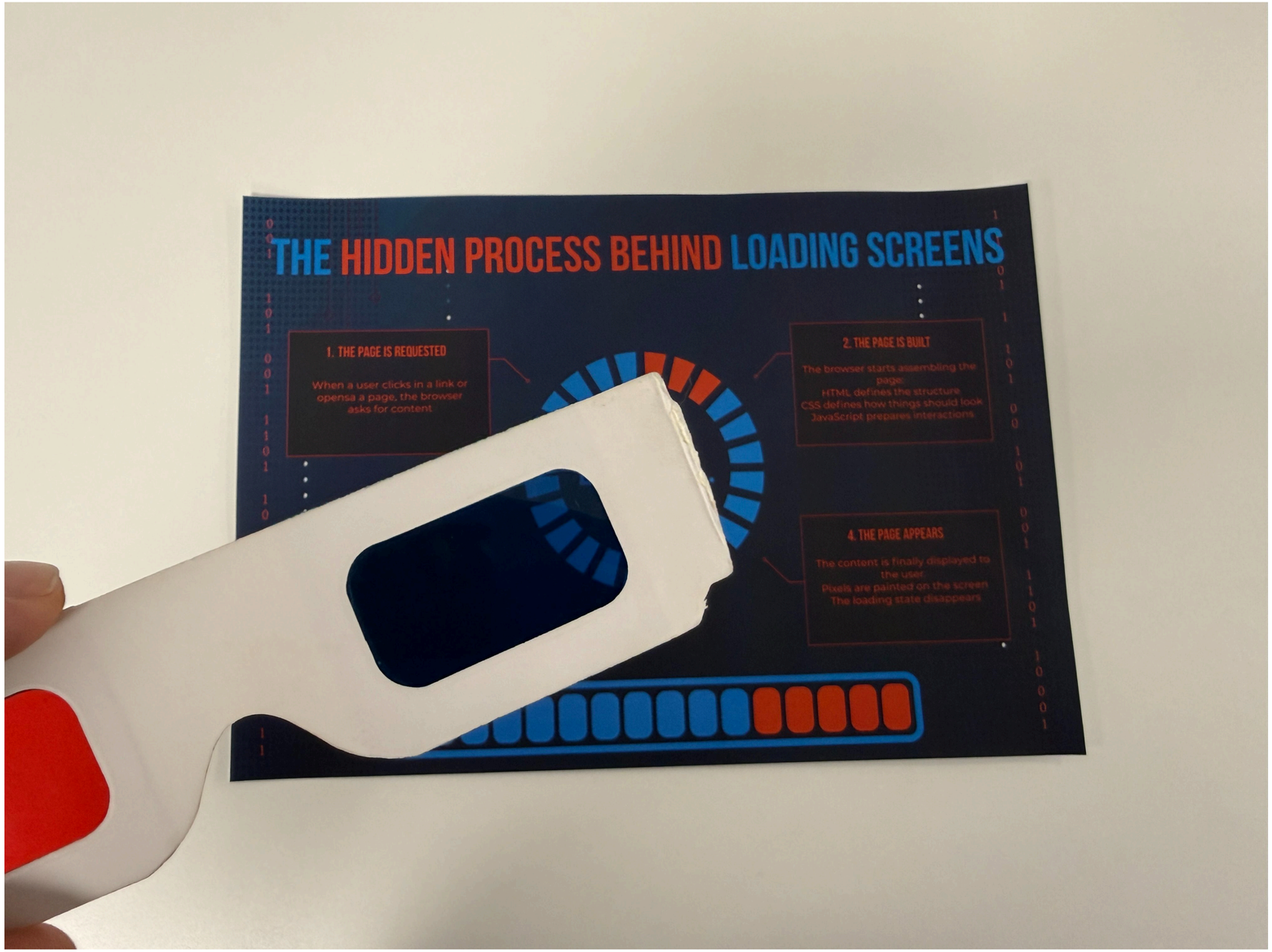


The challenges

The main challenge of this project was the use of colour. In order for the anaglyph effect to function correctly, I was effectively limited to using red and blue for the main content layers. Additionally, the background colours needed to remain very dark in order to maintain clarity and ensure strong contrast between visible and invisible elements.
These challenges influenced many design decisions throughout the project, particularly in terms of hierarchy, balance, and visual legibility.

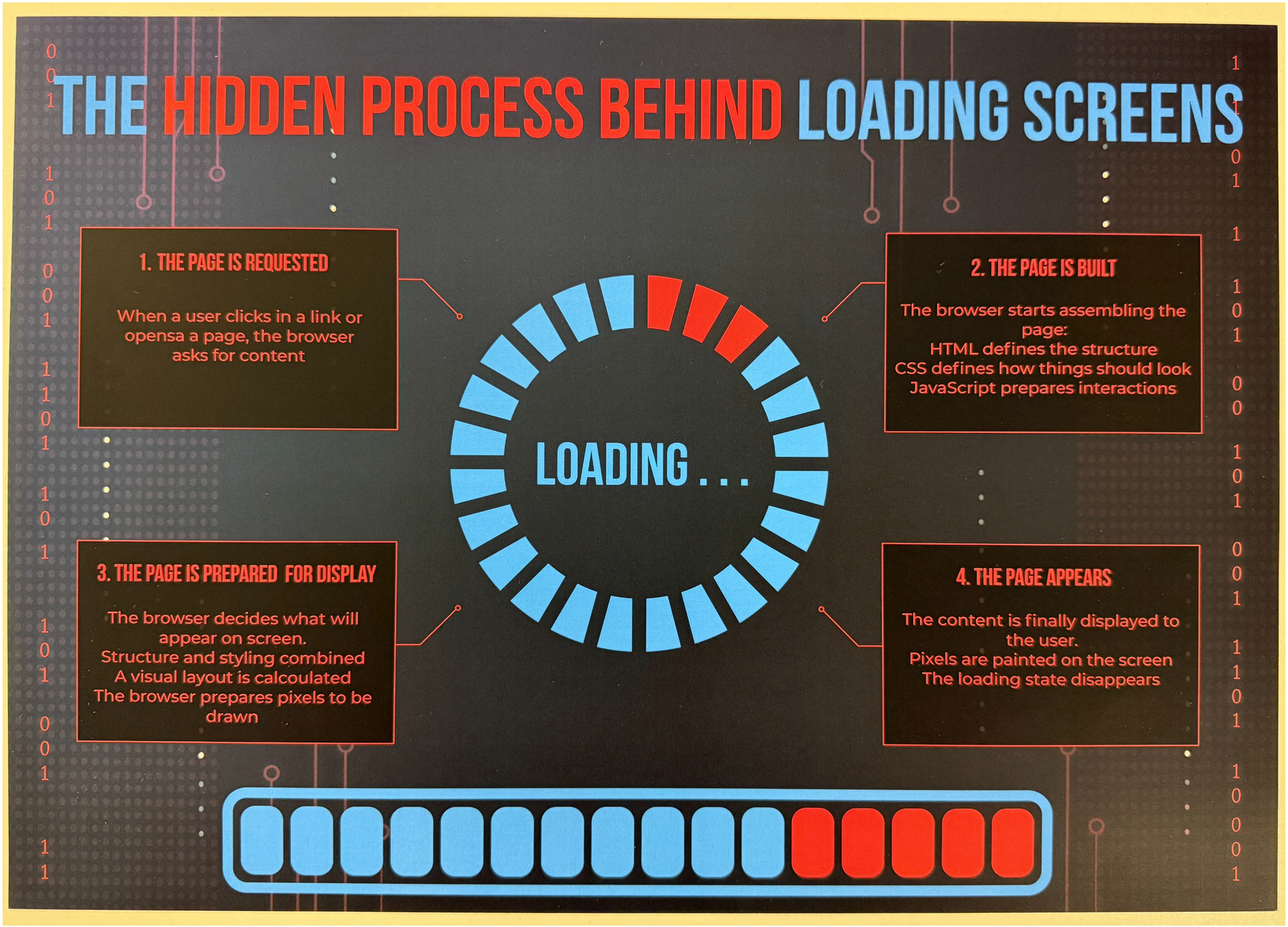


The Project

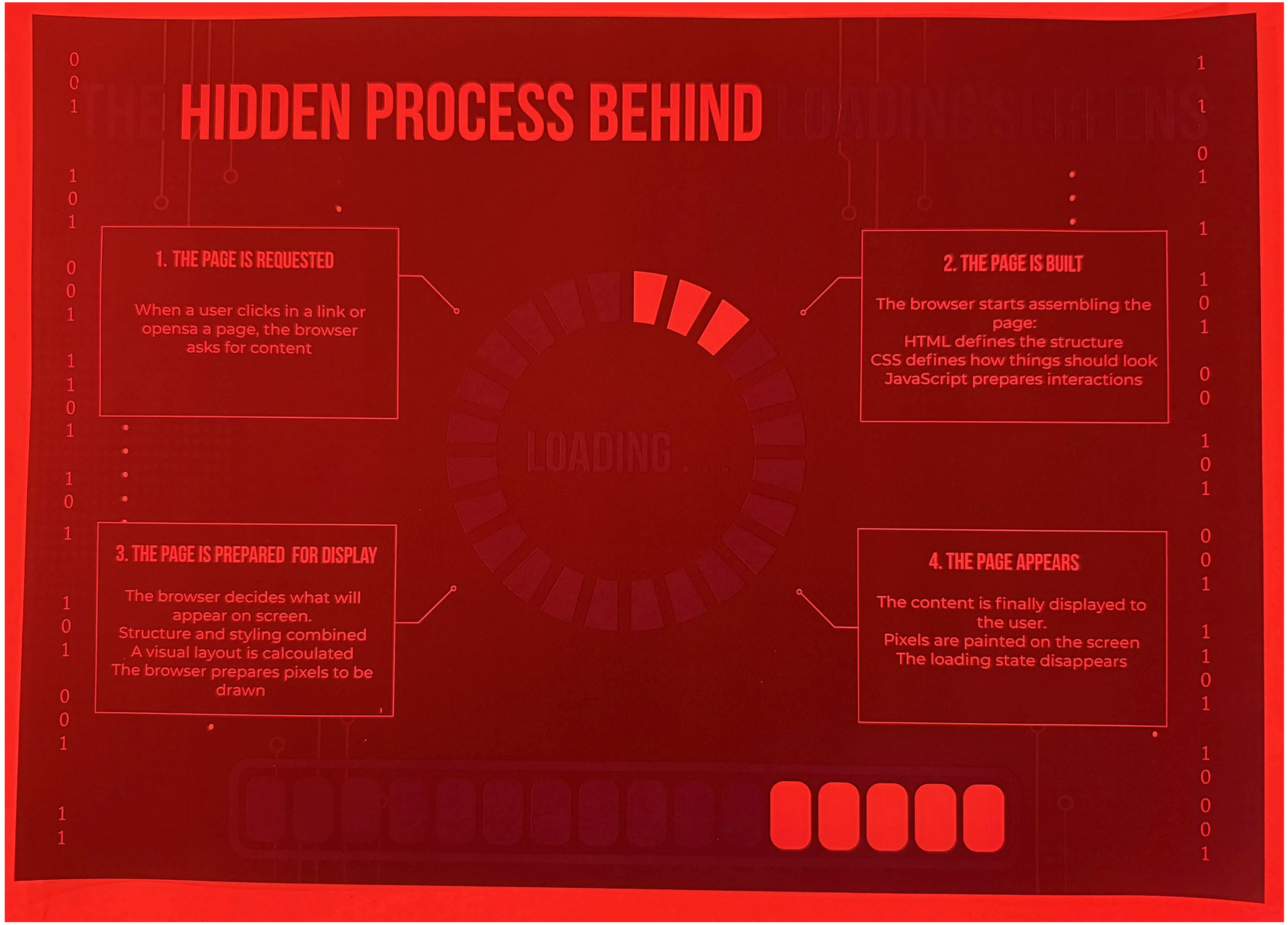

THE HIDDEN PROCESS BEHIND LOADING SCREENS
1. THE PAGE IS REQUESTED
When a user clicks in a link or opensa a page, the browser asks for content
2. THE PAGE IS BUILT
The browser starts assembling the page:
HTML defines the structure
CSS defines how things should look
JavaScript prepares interactions
4. THE PAGE APPEARS
The content is finally displayed to the user.
Pixels are painted on the screen
The loading state disappears

The Project

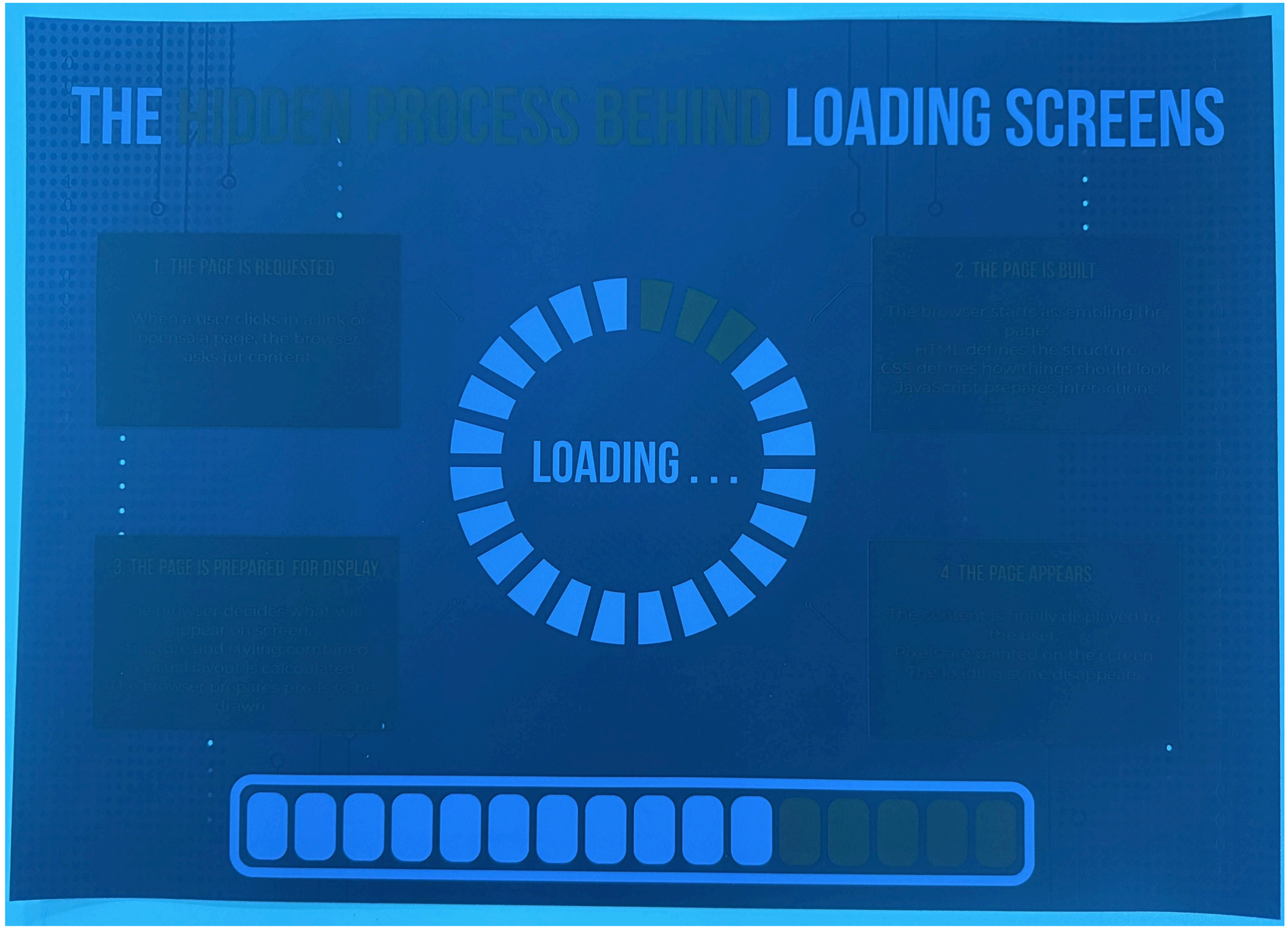

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The hidden processes behind loading pages



The hidden processes behind loading pages



The hidden processes behind loading pages

Information sources and references:

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